On Recovering Latent Factors From Sampling And Firing Graph

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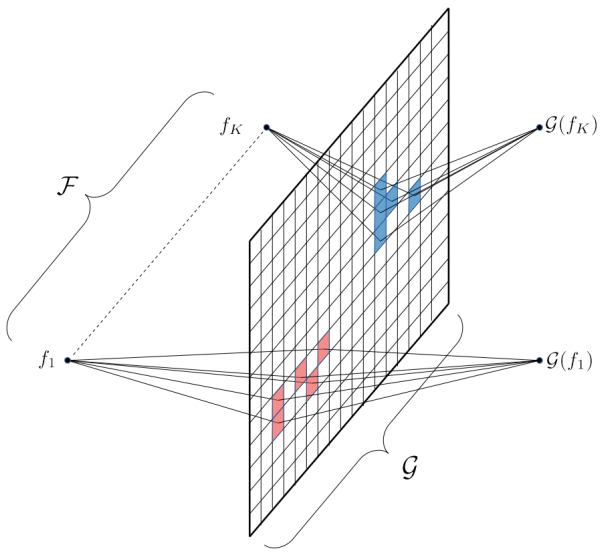
Abstract

Consider a set of latent factors whose observable effect of activation is caught on a measure space that take the form of a grid of bits tacking value in $\{0, 1\}$. This paper intend to deliver a theoretical and practical answer to the question: Given that we have a perfect indicator of the activation of latent factor, is there a procedure that could achieve an optimal identification of factor's activation ?

1 Notations

In order to make the core subject of this report more consistent, we introduce the following notations:

Figure 1: Measure grid model



- $\mathcal{G}$ the measure grid is a set of bit, $\mathcal{G} = \{b_1, \dots b_n\}$.
- n the size of the measure grid, $|\mathcal{G}| = n$.

- $S(\mathcal{G})$ the set of all possible permutation of $\mathcal{G}$, $|S(\mathcal{G})|= 2^n$.
- $S(\mathcal{G}, l)$ the set of all permutation of $\mathcal{G}$ of size l.
- $\mathcal{F}$ the set of latent factor, $\mathcal{F} = \{f_1, \dots f_K\}$.
- K the number of latent factor, $|\mathcal{F}|= K$.
- $S(\mathcal{F})$ the set of all possible permutation of $\mathcal{F}$, $|S(\mathcal{F})|= 2^K$.
- $S(\mathcal{F}, l)$ the set of all permutation of $\mathcal{F}$ of size l.
- $\mathcal{G}(f)$ the set of bits activated by factor f, $\mathcal{G}(f) \in S(\mathcal{G})$ and $f \in \mathcal{F}$.
- $\mathcal{G}^{-1}(b)$ the set of factors that activate grid's bit b, $\mathcal{G}^{-1}(b) \in S(\mathcal{F})$ and $b \in \mathcal{G}$.

2 Definitions And Properties

2.1 Statistical definitions

In this section we provide formalism for the statistical description of factor's activity, there signature on the measure grid as well as the activation estimator.

Activation of factors

Each factor takes value in $\{0, 1\}$ at each instant of time. A factor with value 1 at some instant is active otherwise it has value 0. At this stage of the paper we assume no particular statistical modelisation of factor. Nevertheless if we consider the set of all possible combination of active and unactive factors $(F(2)^K)$, we assume that there is a well defined distribution d' such that

$$d' : F(2)^K \to [0, 1]$$
$$d'_x = \mathbb{P}\left(f_1 = x_1, \dots, f_K = x_K\right)$$

Factors have statistical signatures on the measure grid. That is, there activation lead to the activation of bits of the measure grid. Again, at this stage only the minimum requirement are specified for its statistical modelisation. That is for any $I \in S(\mathcal{G})$

$$\mathbb{P}(\mathcal{G}(f) = I) \in [0, 1]$$
$$\sum_{I \in S(\mathcal{G})} \mathbb{P}(\mathcal{G}(f) = I) = 1$$

The modelisation of latent factor's activation and link to measure grid induce a distribution over combinations of activation of measure grid's bits. We refer to this distribution as d

$$d : F(2)^n \to [0, 1]$$
$$d_x = \mathbb{P}\left(b_1 = x_1, \dots, b_n = x_n\right)$$

Furthermore, for $I \in S(\mathcal{F})$, the modelisation of latent factor to bit' s links verifies also

$$\mathbb{P}(\mathcal{G}^{-1}(b) = I) \in [0,1]$$
$$\sum_{I \in S(\mathcal{F})} \mathbb{P}(\mathcal{G}^{-1}(b) = I) = 1$$

Characteristic polynome

The activity of a factors and grid's bit may be modelized using a set of multivariate polynomials whose fiber and image domain is respectively $F(2)^n$ and $F(2)$. $F(2)$ denotes the fields of two elements, 0 and 1, equipped with the logical XOR and logical AND as the addition and multiplication, respectively. The set of polynome associated with a set $I \in S(\mathcal{F})$ or $I' \in S(\mathcal{G})$ is denoted $\{\mathcal{P}_{I,l}\}_{l \in \mathbb{N}}$ and represents respectively a segmentation of states of factors and measure grid's bits in I or I'. That is, in the measure grid's space space

$$\mathcal{P}_{I',l} : F(2)^n \to F(2)$$
$$\mathcal{P}_{I',l} = \begin{cases} \sum_{\pi \in S(I',l)} x_{\pi_1} \cdot x_{\pi_2} \cdot \ldots \cdot x_{\pi_l}, & \text{if } \ l \in \{1, \ldots, |I|\} \\ 0, & \text{otherwise} \end{cases}$$

Where $S(I', l)$ is the set of all permutation of size l of element of I' and $x_b \in F(2)$ is the state of a bit $b \in \mathcal{G}$. Furthermore we define the characteristic polynomial of the set of bits I' at level $l_0 \in \{0, \ldots, |I'|\}$ as

$$\mathcal{P}^{l_0}_{I'} : F(2)^n \to F(2)$$
$$\mathcal{P}^{l_0}_{I'} = \sum_{l=l_0}^{|I'|} \mathcal{P}_{I',l}$$

The same polynomial can be expressed to modelize states of set of factors. So far, the addition is set to be the logical XOR in the definition of fields $F(2)$. However, in the rest of this report, we will use symbol $+$ and $\sum$ as representation of a logical OR in $F(2)$. This notation enables us to save a lot of time in writing complex polynomials. Denoting $\oplus$ the logical XOR and $\bar{x}$ the opposite of x, one have

$$(x \cdot \bar{y}) \oplus (\bar{x} \cdot y) \oplus (x \cdot y) = x + y$$

Operators on polynome

In order to qualify a set of factors and grid's bits, we define some basic operator. First, let $I_{\mathcal{G}}$ some subset of $S(\mathcal{G})$. Then we denote by $F_2(I_{\mathcal{G}}, \mathcal{G})$ the operators that transform $I_{\mathcal{G}}$ into a set of $|I_{\mathcal{G}}|$ vector in $F(2)^{|\mathcal{G}|}$.

$$F_2(I_{\mathcal{G}}, \mathcal{G}) : S(\mathcal{G}) \to F(2)^{|\mathcal{G}| \times |I_{\mathcal{G}}|}$$

For each of the vector $X \in F_2(I_{\mathcal{G}}, \mathcal{G})$ that has been generated by corresponding set $I \in I_{\mathcal{G}}$, an entry takes value 1 if the associated element belongs to I, 0 otherwise. This oprerator is convenient to evaluate the characteristic polynome of a set of grid's bit. As an example, let $I \in S(\mathcal{G})$ and $l_0 \in \{1, \dots |I|\}$, then

$$\sum_{x \in F_2(\{I'\}, \mathcal{G})} \mathcal{P}_I^{l_0}[x] = \begin{cases} 1, & \text{if } |I' \cap I| \geq l_0 \\ 0, & \text{otherwise} \end{cases}$$

With I' any subset of $\mathcal{G}$. Let the ditribution over states of grid's bits be denoted $d = \{d_x\}_{x \in F_2(S(\mathcal{G}), \mathcal{G})}$ with $d_x = \mathbb{P}(X = x)$, Then the norm of a characteristic Polynome $\mathcal{P}_I^{l_0}$ with respect to distribution d is defined as

$$\begin{aligned} \|.\|_d &: \mathcal{P}_{F_2(S(\mathcal{G}), \mathcal{G})} \to [0, 1] \\ \|\mathcal{P}_I^{l_0}\|_d &= \sum_{x \in F_2(S(\mathcal{G}), \mathcal{G})} \mathcal{P}_I^{l_0}[x] \times d_x \end{aligned}$$

Where $\forall x \in F_2(S(\mathcal{G}), \mathcal{G})$, $\mathcal{P}_{F_2(S(\mathcal{G}), \mathcal{G})}$ the space of all polynomials with domain $F_2(S(\mathcal{G}), \mathcal{G})$ and $\times$ is the simple multiplication in $\mathbb{R}$. Finally, keeping previous notations, let $\{\mathcal{P}_{I_i}^{l_i}\}_{i=1,\dots,k}$ a set of characteristic polynome for some $k \in \mathbb{N}$, we define the product operator as

$$\begin{aligned} \langle ., \dots, . \rangle_d &: \mathcal{P}^k_{F_2(S(\mathcal{G}), \mathcal{G})} \to [0, 1] \\ \langle \mathcal{P}_{I_1}^{l_1}, \dots, \mathcal{P}_{I_k}^{l_k} \rangle_d &= \sum_{x \in F_2(S(\mathcal{G}), \mathcal{G})} \left(\mathcal{P}_{I_1}^{l_1}[x] \cdot \dots \cdot \mathcal{P}_{I_k}^{l_k}[x] \right) \times d_x \end{aligned}$$

Where $\cdot$ denotes the usual multiplication in $F(2)$ and $\times$ is the simple multiplication in $\mathbb{R}$. Every operator specified above in the measure grid space can be defined in the factor space, taking distribution of factor's activation d' and input domain $\mathcal{F}$ instead of distribution d and input space $\mathcal{G}$.

Stochastic processus induced by factor's activation

The activation of factors can be observed through time where at each instant t the realisation of the state of factors follow the distribution d_x and are two-by-two independant. Thus each polynome in the factor space $\mathcal{P}_I^l$ with $I \in S(\mathcal{F})$ and $1 \geq l \geq |I|$ can be observe through time. That is, at each instant $t \in \mathbb{N}$, we define

$$x_I^l[t] = \begin{cases} 1, & \text{with probability } \|\mathcal{P}_I^l\|_{d'} \\ 0, & \text{Otherwise} \end{cases}$$

The stochastic process that takes value 1 if the factor actvivates and 0 otherwise. The activation of factors leads to the activation of grid's bits. Again it induces a stochastic process for each characteristic polynome in the grid's space $\mathcal{P}_I^l$ with $I \in S(\mathcal{G})$ and $1 \geq l \geq |I|$.

$$x_I^l[t] = \begin{cases} 1, & \text{with probability } \|\mathcal{P}_I^l\|_d \\ 0, & \text{Otherwise} \end{cases}$$

The stochastic process that takes value 1 if the bit actvivates and 0 otherwise.

2.2 Firing Graph

In this section we provide a description of the main datastructure used in our solution, the firing graph, as well as basic tools to help its analysis.

Graph specification

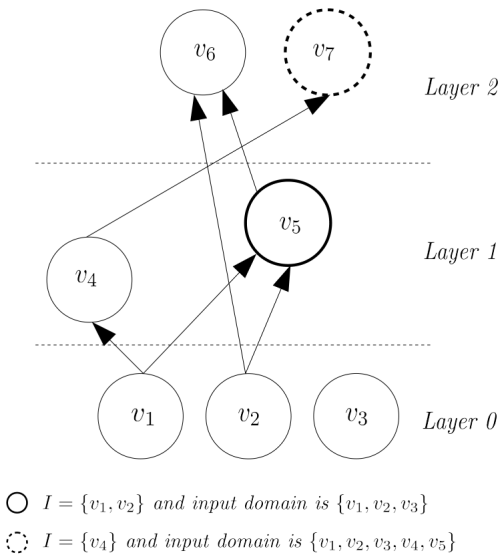
The algorithm presented in this report use a particular datastructure that we refer as firing graph and that we denote $G(V, D_w)$.

- V is the set of vertice $V = \{v_1, \ldots, v_{|V|}\}$
- D_w is the weighted direct link matrix, $D_w \in \mathbb{N}^{|V|\times|V|}$ and $[D_w]_{i,j} = w$ indicate an edge of weight w from vertex v_i to vertex v_j if $w > 0$

G is a directed weighted graph whose vertice are organized in layer. A vertex of some layer $i \in \mathbb{N}$ must have at least one incoming edge from a vertex of layer $i-1$. It may also have outcoming edges toward vertice of layer $j \in \mathbb{N}, j > i$ and incoming edges from vertice of layer $k \in \mathbb{N}, k < i$. Vertice of layer 0 has no incoming edges, they corresponds to bits of the measure grid $\mathcal{G}$. Each vertex stores the tuple (I, l_0)

- I the set of vertice at the tail of incoming edge of the vertex
- l_0 the firing rate's lower bound of the vertex, usually referred as level, $l_0 \in \{1, \ldots, |I|\}$

Figure 2: Firing graph



In the figure we introduce the input domain of a vertex v of the firing graph. It corresponds to the set of vertice that are candidate for an egde toward v. Vertice of the firing graph can also be characterized by there state that can take value 0 or 1 and there propagation through the graph at a rate τ that needs to be set. If at an instant t_0 we set the state of each vertex of layer $0 \leq i < k$ for some $k > 0$, then at instant $t_1 = t_0 + \tau$ vertice of layer k will have there state updated according to their incoming links. The propagation of states through the graph may be modelized by a multivariate boolean polynomial.

Graph Polynomials

As we have previously defined the characteristic polynome of a set of grid's bit and factors, we associate to a vertex $v(I, l_0)$ of the firing graph v the set of polynome $\{\mathcal{P}_{I,l}\}_{l \in \{l_0,\ldots,|I|\ \}}$ that represent a segmentation of the state of a vertex at instant t given the state of vertex in its input domain at instant $t - \tau$. If we denote by n, I and l_0 respectively the size of the input domain of v, the set of vertex that has a link toward v and the level of the vertex, then

$$\mathcal{P}_{v,l} : F(2)^n \to F(2)$$
$$\mathcal{P}_{v,l} = \begin{cases} \sum_{\pi \in S(I,l)} x_{\pi_1} \cdot x_{\pi_2} \cdot \ldots \cdot x_{\pi_l}, & \text{if } \ l \in \{l_0, \ldots, |I|\} \\ 0, & \text{otherwise} \end{cases}$$

Where $S(I, l)$ is the set of all permutation of size l of element of I and for any vertex $v \in V$, $x_v \in F(2)$, is the state of v. Furthermore we define the characteristic polynomial of a vertex v as

$$\mathcal{P}_v : F(2)^n \to F(2)$$
$$\mathcal{P}_v = \sum_{l=l_0}^{|I|} \mathcal{P}_{v,l}$$

Again all operators on polynome defined previously is applicable, let v be some vertex of the firing graph and d_x a ditribution over the combination activation of v's input domain, denoted $d = \{d_x\}_{x \in F_2(S(\mathcal{G}),\mathcal{G})}$ with $d_x = \mathbb{P}(X = x)$. Then the norm of v with respect to distribution d is defined as

$$\|\mathcal{P}_v\|_d = \sum_{x \in F_2(S(\mathcal{G}),\mathcal{G})} \mathcal{P}_v\,[x] \times d_x$$

Similarly, let $v_1, \ldots, v_k$ be any vertice of the firing graph with the same input domain and d_x a ditribution over states of this input domain. Then the product with respect to distribution d is defined as

$$\langle \mathcal{P}_{v_1}, \ldots, \mathcal{P}_{v_k} \rangle_d = \sum_{x \in F_2(S(\mathcal{G}),\mathcal{G})} (\mathcal{P}_{v_1}\,[x] \cdot \ldots \cdot \mathcal{P}_{v_k}\,[x]) \times d_x$$

Finally, the activation of vertices can be seen as strictly stationnary stochastic processus. Given a vertex $v(I, l)$ we define

$$x_v[t] = \begin{cases} 1, & \text{with probability} \quad \|\mathcal{P}_v\|_d \\ 0, & \text{Otherwise} \end{cases}$$

The stochastic process that takes value 1 if the vertex v actvivates and 0 otherwise. d is a well defined distribution over activation of v's input domain.

Connection to grid's bit

The firing graph is a convenient data structure to evaluate polynome of group of measure grid's bits. Let G be a firing graph with its layer 0 composed of the set of bits $\mathcal{G}$, then for any vertext of layer 1 $v(I, |I|)$ the characteristic polynome v is equal to the characteristic polynome of the set of bits $I \in \mathcal{G}$ with level $|I|$.

$$\mathcal{P}_v = \mathcal{P}_{I,|I|}$$
$$x_v[t] = x_I^{|I|}[t-1]$$

Furthermore if we set the level of v to 1 its characteristic polynome become the logical *or*-sum of the characteristic polynome of each bits of I

$$\mathcal{P}_v = \sum_{b \in I} \mathcal{P}_{\{b\},1}$$
$$x_v[t] = \begin{cases} 1, & \text{if} \quad \sum_{b \in I} x_{\{b\}}^1[t-1] > 0 \\ 0, & \text{Otherwise} \end{cases}$$

Besides, one can design more complexed arrangement of vertice that enables to modelize multiple set of measure grid's bits. Let G be a firing graph with its layer 0 composed of the set of bits $\mathcal{G}$, let $u(I, |I|)$ and $v(I', 1)$, such that $I \cap I' = \emptyset$, be vertice of layer 1 and $w(\{u, v\}, 2)$ a vertex of layer 2. Then one can easily see that that characteristic polynome of w verifies

$$\mathcal{P}_w = \sum_{b \in I'} \mathcal{P}_{I \cup \{b\},|I|+1}$$
$$x_w[t] = x_I^{|I|}[t-1]$$

Finally, if stochastic processus induced by each of the as for measure grid and factor space polynome, the activation of vertice's characteristic polynomes can be observed through time and induce stochastic processus at each vertex of the graph.

2.3 Evaluation of measure grid's bits

A perfect indicator of the activation of a given factor f can be used as target activation behaviour to evaluate the possibility of any set of bits to be part of the signature of the factor on the measure grid.

Factor's signature

The activation of factors can be modeled by a polynome in the factor space defined as

$$\mathcal{P}_f : F(2)^K \to F(2)$$
$$\mathcal{P}_f = \mathcal{P}_{\{f\},1}$$

Further we could also describe the activity of a factor f by a polynome in the measure grid's space

$$\mathcal{P}_{\mathcal{G}(f)} : F(2)^n \to F(2)$$
$$\mathcal{P}_{\mathcal{G}(f)} = \mathcal{P}_{\mathcal{G}(f),|\mathcal{G}(f)|}$$

The characteristic polynomes of a factor $f \in \mathcal{F}$ verifies $\forall\ I \subset \mathcal{F}$ and $I' = \bigcup_{f' \in I} \mathcal{G}(f')$ let $x, x' \in F(2)^K \times F(2)^n$ such that $x \in F_2(\{I\}, \mathcal{F})$ and $x' \in F_2(\{I'\}, \mathcal{G})$

$$\mathcal{P}_f[x] = 1 \Rightarrow \mathcal{P}_{\mathcal{G}(f)}[x'] = 1$$

Furthermore if $!\,\exists J \in \mathcal{F} \setminus \{f\}$ such that $\mathcal{G}(f) \subset \bigcup_{f' \in J} \mathcal{G}(f')$

$$\mathcal{P}_f[x] = 1 \Leftrightarrow \mathcal{P}_{\mathcal{G}(f)}[x'] = 1$$

basic metrics

Let $I \in S(\mathcal{G})$, $l \in \{1, \ldots, |I|\}$, $f \in \mathcal{F}$ and e be respectively some measure grid's bit, some level, some latent factor and the event "factor f is active". Then we define a first metrics, refered as recall coefficient, defined as

$$\mu_{I,l,f} = \langle \mathcal{P}_I^l, \mathcal{P}_{\mathcal{G}(f)} \rangle_{d|e} + \langle \mathcal{P}_I^l, \mathcal{P}_{\mathcal{G}(f)}^{\perp} \rangle_{d|e}$$

Where $d|e$ is the distribution over measure grid's bits activation given event e and $\mathcal{P}_{\mathcal{G}(f)}^{\perp}$ is the complement of $\mathcal{P}_{\mathcal{G}(f)}$ in $F(2)$. Furthermore we define the precision coefficient as

$$\nu_{I,l,f} = \langle \mathcal{P}_I^l, \mathcal{P}_{\mathcal{G}(f)} \rangle_{d|\bar{e}} + \langle \mathcal{P}_I^l, \mathcal{P}_{\mathcal{G}(f)}^{\perp} \rangle_{d|\bar{e}}$$

Where $d|\bar{e}$ is the distribution over measure grid's bits activation given not event e. Finally we define the purity coefficient as

$$\omega_{I,l,f} = \frac{\nu_{I,l,f}}{\mu_{I,l,f}}$$

The lower $\omega_{I,l,f}$ the more pure is the set I with respect to f. Furthermore one can easily see that for any set $I \in S(\mathcal{G})$ we have

$$\omega_{I,l,f} \geq \omega_{\mathcal{G}(f),|\mathcal{G}(f)|,f}$$

Finally the recall, precision and purity coefficient can be defined for any vertex v of a firing graph where vertex of layer 0 are composed by measure grid's bit and are denoted respectively $\mu_{v,f}$, $\nu_{v,f}$ and $\omega_{v,f}$.

advanced metrics

Let $I \in S(\mathcal{G})$, $l \in \{1, \ldots, |I|\}$, $f \in \mathcal{F}$ and e be respectively some measure grid's bit, some integer, some latent factor and the event "factor f is active". Then we define the precision of the set I with respect to factor f at level l_0 as

$$\phi_{I,l,f} = \frac{\|\mathcal{P}_I^l\|_{d,e}}{\|\mathcal{P}_I^l\|_d}$$

We also define the recall of the set I with respect to factor f at level l_0 as

$$\psi_{I,l,f} = \frac{\|\mathcal{P}_I^l\|_{d,e}}{\|\mathcal{P}_{\mathcal{G}(f)}\|_{d,e}}$$

Where d, e is a well defined distribution over combination of activation of measure grid's bits that intetersect with event e. Finally the precision and the metrics can be defined for any vertex v of a firing graph where vertex of layer 0 are composed by measure grid's bit and are denoted respectively $\phi_{v,f}$ and $\psi_{v,f}$.

Advanced stochastic process induced by vertex

Given that the layer 0 of the firing graph is compose of measure grid's bits, the propagation of activation through the firing graph induces multiple stochastic processus in each of its vertex. Given a vertex v at layer $k \geq 0$, its characteristic polynome $\mathcal{P}_v$, a factor f and its characteristic polynome in the measure grid space $\mathcal{P}_f$ the activity of the node is described by

- The reward process with respect to factor f, $r_{v,f}[t]$, $t \in \mathbb{N}$

  $r_{v,f}[t] = r_{v,f}[t-\tau] + r_{v,t,f}$, with $\{r_{v,t,f}\}_{t\in\mathbb{N}}$ a set of i.i.d random variable that takes value 1 if vertex v activate at instant t and the event "factor f activate" was true at time $t - k \times \tau$. That is, $\forall\ t < k \times \tau$, $r_{v,t,f} = 0$ and $\forall\ t \geq k \times \tau$

  $$r_{v,t,f} = \begin{cases} 1, & \text{with probability } q_r = \|\mathcal{P}_v\|_{d,e} \\ 0, & \text{with probability } 1 - q_r \end{cases}$$

  where d, e is the distribution over the combination of activation of vertex of layer 0 that intersect with the event e: "factor f is active".

- The penalty process with respect to factor f, $p_{v,f}[t]$, $t \in \mathbb{N}$

  $p_{v,f}[t] = p_{v,f}[t-\tau] + p_{v,t,f}$, with $\{p_{v,t,f}\}_{t\in\mathbb{N}}$ a set of i.i.d random variable that takes value 1 if vertex v activate at instant t and the event "factor f activate" was false at time $t - k \times \tau$. That is, $\forall\ t < k \times \tau$, $p_{v,t,f} = 0$ and $\forall\ t \geq k \times \tau$

$$p_{v,t,f} = \begin{cases} 1, & \text{with probability } q_p = q_n - q_r \\ 0, & \text{with probability } 1 - q_p \end{cases}$$

- The score process with respect to factor f, $s_{v,f}\left[N, T, \rho\right]$, $N, T, \rho \in \mathbb{N}^3$

  $s_{v,f}\left[N, T, \rho\right] = N + \sum_{t=1}^{T} s_{v,\rho,t,f}$, with $\{s_{v,\rho,t,f}\}_{t\in\mathbb{N}}$ a set of i.i.d random variable. $s_{v,\rho,t,f}$ takes value 1 if the event "factor f activate" was true at time $t - k \times \tau$ and value $-\rho$ if the event was false, given that v activate at t. That is, $\forall\, t < k\times\tau$, $p_{v,t,f} = 0$ and $\forall\, t \geq k\times\tau$

$$s_{v,\rho,t,f} = \begin{cases} 1, & \text{with probability } q_s = \frac{q_r}{q_r+q_p} \\ -\rho, & \text{with probability } 1 - q_s \end{cases}$$

The stochastic processus of each vertex is a convenient way to estimate the evaluation metrics of sets of measure grids bit's whose behaviour is caught be a vertex of the firing graph.

2.4 Properties

This paragraph intend to deliver useful properties for the analysis of the algorithm. The proof of every properties can be found in the appendice at the end of this paper.

Polynomial decomposition

<u>Partition</u>

Let $v_1(I, l_0)$, $v_2(J, 0)$ and $v_3(K, 0)$, be three vertice of the same firing graph, with the same input domain $\mathcal{G}$ such that $I = J \cup K$ and $J \cap K = \emptyset$, then, $\forall x \in F_2(S(\mathcal{G}), \mathcal{G})$

$$\mathcal{P}_I^{l_0}\left[x\right] = \sum_{l=l_0}^{|I|} \sum_{j=0}^{|J|} \mathcal{P}_{J,j}\left[x\right] \cdot \mathcal{P}_{K,l-j}\left[x\right] \tag{1}$$

In paticular for $v \in I$

$$\mathcal{P}_{I,l}\left[x\right] = \mathcal{P}_{I\setminus\{v\},l}\left[x\right] \cdot \mathcal{P}_{\{v\},0}\left[x\right] + \mathcal{P}_{I\setminus\{v\},l-1}\left[x\right] \cdot \mathcal{P}_{\{v\},1}\left[x\right] \tag{2}$$

<u>Decomposition</u>

Let G be a firing graph composed of a set of $\mathcal{G}$ of vertice of layer 0. $u(I, l_u)$, $v(I', l_v)$ such that $I \cap I' = \emptyset$ as vertice of layer 1 and $w(\{u, v\}, 2)$ as vertex of layer 2. Let $K \in S(I', l_v)$, $x \in F_2(S(\mathcal{G}), \mathcal{G})$ and $x' = \begin{bmatrix} \mathcal{P}_u[x] & \mathcal{P}_v[x] \end{bmatrix}$ then

$$\mathcal{P}_{K,|K|}\left[x\right] \cdot \mathcal{P}_{\{u,v\},2}\left[x'\right] = \sum_{l=l_u}^{|I|} \sum_{J\in S(I,l)} \mathcal{P}_{J\cup K,l+|K|}\left[x\right] \tag{3}$$

In particular if $l_u = |I|$ and $l_v = 1$, then for any vertex of layer 0 $b \in I'$

$$\mathcal{P}_{b,1}\left[x\right] \cdot \mathcal{P}_{\{u,v\},2}\left[x'\right] = \mathcal{P}_{I\cup\{b\},|I|+1}\left[x\right] \tag{4}$$

Metrics

Throughout this section, we consider G to be a firing graph with layer 0 composed by measure grid's bits $\mathcal{G}$ and $f \in \mathcal{F}$ denote some target factor that is linked to some bit of the measure grid. The distribution of activation of latent factors and measure grid's bits will be denoted d and d' and the event "factor f is active" will be denoted by e.

<u>Precision of vertex</u>

Let v be a vertex connected to a set I of measure grid's bits with level $l \in \{1, \ldots, |I|\}$ and let $f \in \mathcal{F}$ be some factor. Then the precision of v with respect to f is

$$\phi_{v,f} = \frac{\|\mathcal{P}_f\|_{d'}}{\|\mathcal{P}_f\|_{d'} + (1 - \|\mathcal{P}_f\|_{d'}) \times \omega_{v,f}} \tag{5}$$

Furthermore, if $\mu_{v,f} = 1$ we have

$$\phi_{v,f} \leq \frac{\|\mathcal{P}_f\|_{d'}}{\|\mathcal{P}_f\|_{d'} + (1 - \|\mathcal{P}_f\|_{d'}) \times \omega_{\mathcal{G}(f),|\mathcal{G}(f)|,f}} \tag{6}$$

<u>Recall of vertex</u>

Let v be a vertex connected to a set I of measure grid's bits with level $l \in \{1, \ldots, |I|\}$ and let $f \in \mathcal{F}$ be some factor. Then the recall of v with respect to f is

$$\psi_{v,f} = \mu_{v,f} \tag{7}$$

Furthermore,

$$0 \leq \phi_{v,f} \leq 1 \tag{8}$$

Where right equality is reached whenever v is connected to a set of measure grid's bit $I \in \mathcal{G}$, with level $l_0 = |I|$ such that $I \subset \mathcal{G}(f)$.

stochastic processus

Again, throughout this section, we consider a firing graph G where the layer 0 is composed by measure grid's bits. v denote some vertex of G and $f\mathcal{F}$ denote some factor. Finally we use d and d' to respectively denote the distribution over combination of activation of measure grid's bits and factors and e to denote the event "factor f is active".

<u>vertex's reward process</u>

If denotes penalty process of v with respect to f as defined in section 1.2.2, then

$$\mathbb{E}\left[r_{v,f}[t]\right] = t \times q_r = t \times (\mu_{v,f} \times \|\mathcal{P}_f\|_{d'}) \tag{9}$$

vertex's penalty process

if $p_{v,f}[t]$ denotes penalty process of v with respect to f as defined in section 1.2.2, then

$$\mathbb{E}\left[p_{v,f}[t]\right] = t \times q_p = t \times (\nu_{I,f} \times (1 - \|\mathcal{P}_f\|_{d'})) \tag{10}$$

vertex's score process

If $s_{v,f}[N, T, \rho]$ denotes the score process of v with respect to f as defined in section 1.2.2 with some $N, T, \rho \in \mathbb{N}^3$, then

$$\mathbb{E}\left[s_{v,f}[N, T, \rho]\right] = N + T \times (q_s \times (\rho + 1) - \rho) = N + T \times (\phi_{v,f} \times (\rho + 1) - \rho) \tag{11}$$

Furthermore,

$$\mathrm{Var}\left[s_{v,\rho,t,f}\right] = (1 + \rho)^2 \times \phi_{v,f} \times (1 - \phi_{v,f}) \tag{12}$$

3 Identification of Latent Factor

In this section, we presents the a couple of operations that can be used for the identification of a latent factor:

- Sampling: Sampling the measure grid and build a firing graph
- Draining: Draining vertex of the firing graph keeping only low purity coefficient's vertex

At this stage both processus will be described and the efficiency of the draining algorithm quantified.

3.1 Sampling

The sampling of the measure grid is a specific procedure followed in order to select some bits of the grid. This procedure is usually designed to be the most efficient in the fullfilment of specific quantitative objective. First we assume that given a target factor f we have access to a determinist exact indicator of f's activation. Then given target f, the objective of our sampling is to maximize the probability that we sample a bit b whose purity coefficient with respect to f is higher or equal to some positive constant ω. That is, if we denote by s the random variable of the outcome of a single sampling, then the objective is to maximize

$$\mathbb{P}(\omega_{\{s\},1,f} \geq \omega)$$

Again if we have a set of $I \subset \mathcal{G}$ bits, a target factor f, then the objective of sampling is to maximize the probability of sampling a bit b for which the purity of set $I \cup \{b\}$ is higher or equal to a given positive constant ω. That is, if we denote by s the random variable of the outcome of a single sampling, then the objective is to maximize

$$\mathbb{P}(\omega_{I\cup\{s\},|I|,f} \geq \nu)$$

We propose a very intuitive sampling method based on the indicator of activation of target factor. Given parameters $p_s \in [0,1]$ and S_p respectively the probability of picking a bit and a set of preselected measure grid's bits, the sampling proceudre writes

Algorithm 1 Sampling

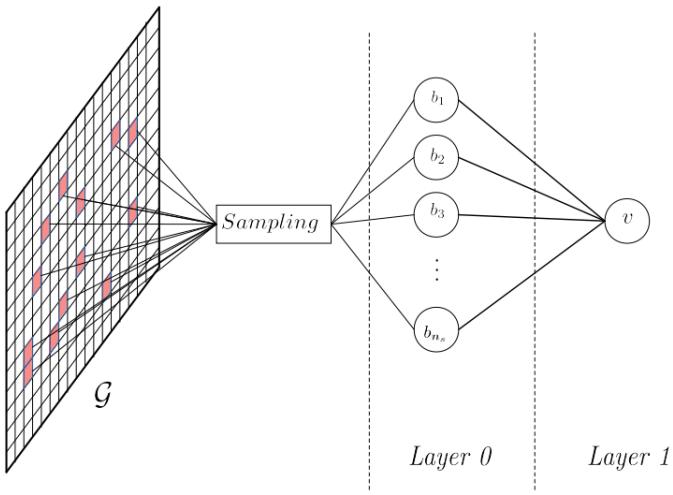
Input: $p_{\mathcal{S}}$, S_p

Output: S

```
S ← {}, x_f ← nextFactorState(), X_G ← nextGridState()
while S == {} do
    if x_f == 1 and ∀b ∈ S_p x_b == 1 then
        for all b ∈ G \ S do
            if X_G[b] == 1 then
                S ← S ∪ {b} with probability p_s
            end if
        end for
    end if
    x_f ← nextFactorState()
    X_G ← nextGridState()
end while
```

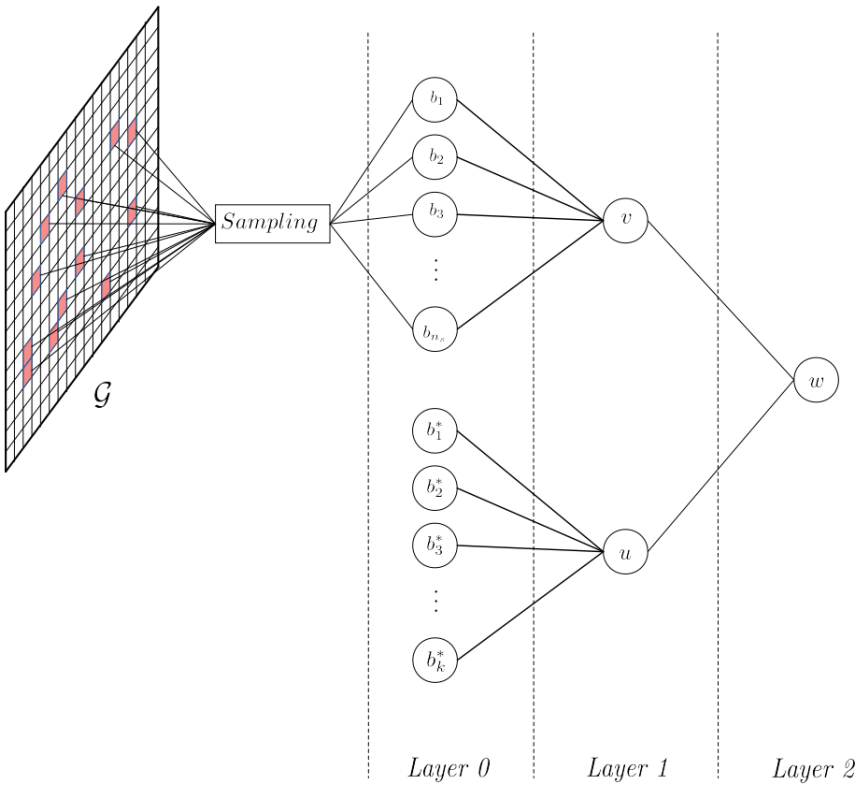
Where x_f and $X_{\mathcal{G}}$ are respectively a scalar that take value 1 when factor f is active, 0 otherwise and a mapping with measure grid's bits as keys and their state as value (0 or 1). The second mean of the sampling procedure is to build a firing graph from which the evaluation of the purity coefficient of samples can be estimated through the observation of edge's score processus. The construction of the firing graph requires to set a parameter N that corresponds to the initial weigths of edges that will be drained. In addition we add mask matrix $G_{mask} \in \{0,1\}^{|V|}$ that controls wich edges is candidate for draining. We classify firing graph into two classes

Figure 3: Single sampled firing graph



In this first case sampled bits $\{b_1, \ldots, b_{n_s}\}$ are used as vertex of the layer 0 of a firing graph G. Then the vertex $v(\{b_1, \ldots, b_{n_s}\}, 1)$ is added at the layer 1 of G. In this case with set G_{mask} so that only layer 0's outcoming edges are candidate for draining.

Figure 4: Joint sampled firing graph



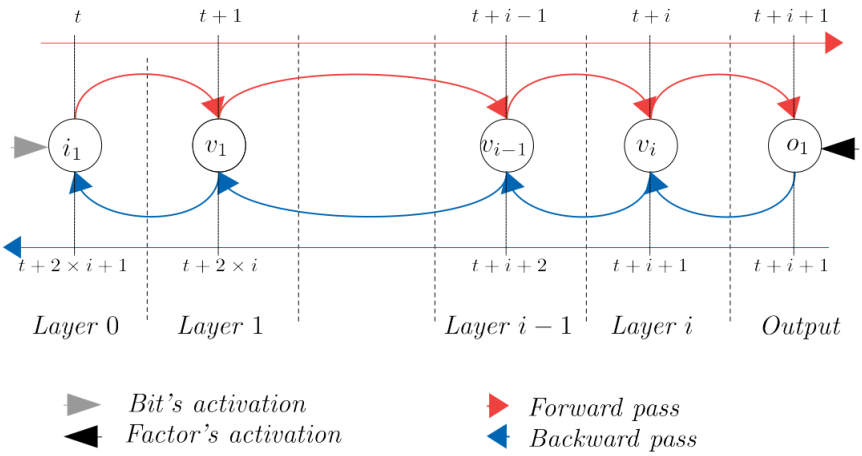
In this second case sampled bits $\{b_1, \ldots, b_{n_s}\}$ and the pre-selected bits $\{b_1^*, \ldots, b_i^*\}$ for some $i \in \mathbb{N}^*$ compound the layer 0. Then the vertices $v(\{b_1, \ldots, b_{n_s}\}, 1)$ and $u(\{b_1, \ldots, b_{n_s}\}, k)$ are

added to layer 1 of G with outcoming edge's weights set to ∞ . Finally, the vertex $w(\{u, v\}, 2)$ is added at layer 2 of G. In this case with set G_{mask} so that only $b_1, \ldots, b_{n_s}$'s outcoming edges are candidate for draining.

3.2 Draining

Draining the firing graph built during the sampling pass consists in iterating a forward propagation of bit's activation through firing graph and backward propagation of feedback generated by factor's activation. Feedback are meant to increment or decrement the weight of an edge. Given that an edge with a null or negative weigth vanishhes, at the end of the routine, the topology of the graph differentiate between the purity of each vertex. To ease understanding of the algorithm we split the vertices of the firing graph into input and core vertices which are respectively vertices of layer 0 and vertices of layer > 0. Furthermore we introduce a new type of vertices that can only have incoming edges from core vertices. We refer to those nodes as output vertices.

Figure 5: Draining diagram



We use n_i, n_c and n_o to refer to the number of respectively input, core and output vertices. Furthermore we define $I \in \mathbb{N}^{n_i \times n_c}$, $C \in \mathbb{N}^{n_c \times n_c}$ and $O \in \mathbb{N}^{n_c \times n_o}$ that respectively corresponds to the weighted direct link matrices respectively from input toward core vertices, core toward core vertices and core toward output vertices. As we have seen in previous section, the activation of grid's bits can be observed as stochatic processus that takes value $\{0, 1\}$ that propagates through graph and induced core vertice's stochastic processus. Thus we define the following stochastics vectors

- $x_i^{(t)} \in \{0, 1\}^{1 \times n_i}$ the vector of activation of input vertices at instant t
- $x_c^{(t)} \in \{0, 1\}^{1 \times n_c}$ the vector of activation of core vertices at instant t
- $x_o^{(t)} \in \{0, 1\}^{1 \times n_o}$ the vector of activation of output vertices at instant t

The propagation of activation through the firing graph can be represented with two equations:

Forward transmitting (FT)

$$\tilde{x}_c^{(t)} = x_i^{(t-1)} \cdot I + x_c^{(t-1)} \cdot C$$
$$\tilde{x}_o^{(t)} = x_c^{(t-1)} \cdot O$$

Forward processing (FP)

$$[x_c^{(t)}]_i = \begin{cases} 1, & \text{if } [\tilde{x}_c^{(t)}]_i > l_i \\ 0, & \text{Otherwise} \end{cases}$$
$$[x_o^{(t)}]_j = \begin{cases} 1, & \text{if } [\tilde{x}_o^{(t)}]_j > 1 \\ 0, & \text{Otherwise} \end{cases}$$

Where $\cdot$ is the usual dot product, $(i, j) \in \{1, \dots n_c\} \times \{1, \dots n_o\}$ and l_i is the level of the ith core vertex. An output vertex of the firing graph is feeded with the activation of a targeted factors decayed in time that is usually chosen to be the number of layer - 1. For single and joint sampled firing graph the decay is respectively set to 1 and 2. The factor activation enables to generate a feedback to the output that is back propagated through the firing graph. Supposing that we set the factor's decay to $d \geq 1$, the feedback is defined as

$$x_{b,o}^{(t)} = x_o^{(t)} \circ \left((\rho + 1) \times x_f^{(t-d)} - \rho \right)$$

Where $\circ$ denotes the usual Hadamard product and $x_f^{(t-d)}$ is the vector of states of factors at instant $t - d$. Backpropagating $x_{b,o}^{(t)}$ correctly up to the input vertices is made possible by using time synchronisation and topology of the firing graph. In order to describe the process we introduce the notion of depths of a vertex. A vertex depths correspond to the set of its distances, in terms of vertex, from any output vertices. We denote by V_i, $i \in \mathbb{N}$ the set of vertices that has a depth of i. Let G be a firing graph with $k \in \mathbb{N}^*$ layers augmented with a layer of ouptut vertices. Let V_o the set of output vertices, $\forall v, o \in V_0 \times V_o$, v is elligible to o's feedback at instant t if and only if

- v was active at instant $t - 1$
- v has an edge toward o

The same principle can be used to backpropagate the feedback from the vetices of V_0 toward vertices of V_1. Generally speaking, the back propagationfrom vertices of V_i toward V_{i+1} respect $\forall v, v' \in V_i \times V_{i-1}$, v is elligible to feedback of v' at instant t if and only if

- v was active at instant $t - (2 \times i + 1)$
- v has an edge toward v'

Finally we can encode the backpropagation equations as

Backward transmitting (BT)

$$\tilde{X}_{b,c}^{(t)} = (O \cdot X_{b,o}^{(t-1)} + C \cdot X_{b,c}^{(t-1)}) \circ X_{m,c}^{(t)T}$$
$$X_{b,i}^{(t)} = (I \cdot X_{b,c}^{(t-1)}) \circ X_{m,i}^{(t)T}$$

Backward processing (BP)

$$X_{b,o}^{(t)} = \begin{bmatrix} \mathbf{0}_{no\times 1} & x_{b,o}^{(t)} & \mathbf{0}_{no\times d_{max}-2} \end{bmatrix}$$
$$X_{b,c}^{(t)} = \begin{bmatrix} \mathbf{0}_{n_c\times 2} & [\tilde{X}_{b,c}^{(t)}]_{(:n_c,:(d_{max}-2))} \end{bmatrix}$$

Structure udpates (SU)

$$O = O \circ (X_{b,o}^{(t-1)} \cdot X_{m,c}^{(t)})^T$$
$$C = C \circ (X_{b,c}^{(t-1)} \cdot X_{m,c}^{(t)})^T$$
$$O = I \circ (X_{b,c}^{(t-1)} \cdot X_{m,i}^{(t)})^T$$

Where $X_{m,c}^{(t)} = \begin{bmatrix} x_c^{(t)} & \dots & x_c^{(t-d_{max})} \end{bmatrix}^T$ and $X_{m,i}^{(t)} = \begin{bmatrix} x_i^{(t)} & \dots & x_i^{(t-d_{max})} \end{bmatrix}^T$, $X_{c,b}^{(t)} \in \{0, 1, -\rho\}^{n_c \times d_{max}}$ for $t \in \mathbb{N}^*$ and $X_{c,b}^{(0)} = \mathbf{0}_{n_c\times d_{max}}$. Furthermore $d_{max} \geq (l-1) \times 2 + 1$ where l is the number of layer of the firing graph. Finally we provide a parameter T to the draining algorithm. It controls the targeted number of feedback that an edge should received before disabling its update. Maintaining update's permissions for each edges requires an operation similar to structure update. Iterating throught the transmit and process operation until either G is composed of two distinct connexe component or no structure update is enabled. At the convergence of the draining algorithm the value of drained edges e is the value of score process $s_{e,f}[N, T, \rho]$.

Algorithm 2 Draining

Input: G, T, T_{max}, ρ, decay
Output: G drained

```
i ← 0                                                    ▷ Initialisation
X_{b,c}, x_{b,o}, x_i, x_c, X_{m,c}, X_{m,i} ← InitSignals()
while i < T_max do                                       ▷ Core loop
    x_i ← nextGridState()
    x_c, x_o ← FT(G, x_i, x_c)                           ▷ Forward pass
    X_{m,c}, X_{m,i}, x_c, x_o ← FP(x_c, x_o)
    if i ≥ decay then
        x_f ← nextFactoreState()
        X_{b,c}, X_{b,o} ← BP(X_{b,c}, X_{b,o}x_f, ρ)    ▷ Backward pass
        G' ← SU(T, G, X_{b,c}, X_{b,o}, X_{m,c}, X_{m,i})
        X_{b,c}, X_{b,i} ← BT(G, X_{b,c}, X_{b,o}, X_{m,c}, X_{m,i},)
        G ← G'
    end if
    if G.cc == 2 or not G_mask.any() then                ▷ Stop conditions
        break
    end if
    i ← i + 1
end while
```

Clearly, the complexity of the algorithm is dominated by the backward transmit and structure update operations. A standard worst case analysis of those operations gives $\mathcal{O}(n^4 \times d_{max}^2)$. Yet this analysis relies on standard complexity time for dense matrix operations, and doesn't take into account sparsity of signals and direct link matrices or the randomness of inputs. In practice we have found that the forward and backward propagation of the bits and factors activation is time consuming, especially when both N and T are large numbers. As a consequence, batch_size successive bits and factor's state are forward and backward propagated through the graph at each iteration. The decrease in time complexity of this practical trick is impressive and worth the slight gain in space comlexity of the algorithm even though it does not stick to the expectation of theoretical worst case analysis. Finally this trick requires to dynamically change the batch_size so that treshold for the number of updates at each edges is respected.

3.3 Analysis of the algorithm

Theorem 3.1 *Given a set of sampled bits S, a set of pre-selected bits $I = \{b_1^*, \dots, b_i^*\}$ a target factor f and G the firing graph built after sampling algorithm with parameter N. There exists a 4-tuple (ω, N, T, ρ) such that the probability of event E: "no input vertices of G have outcoming edges at the end of the draining" is upper bounded. More specifically*

$$\mathbb{P}(E) \leq \sum_{j=0}^{|S|} p^j \times \mathbb{P}_{\mathcal{S}}\left(|\{s \in S \setminus \omega_{I\cup\{s\},|I|+1,f} < \omega\}| = j\right)$$

Where $p = \mathbb{P}\left(s_{v,f}(N,T,\rho) < 0 | \omega_{I\cup\{s\},i+1,f} < \omega_i\right)$, with $\{b_1^*, \dots, b_i^*, s\}$ compose the layer 0 and $v(I \cup \{s\}, i+1)$ the layer 1 of a firing graph of 2 layers, Furthermore

$$\mathbb{P}\left(s_{v,f}(N,T,\rho < 0) | \omega_{I\cup\{s\},i+1,f} < \omega\right) \leq C \times \max\left(\exp\left(-T \times \left(\frac{\delta^+ \delta_{i,f} c}{\sigma}\right)^2\right), \exp\left(-T \times \delta^+ \delta_{i,f} c\right)\right)$$

With $\delta_{i,f}$, δ^+ C and c are postitive constants that depends on ω and i and $\mathrm{Var}[s_{v,\rho,t}] = \sigma^2$.

Proof. As a reminder, in the core of this proof we refer to d and d' respectively to the distribution over bit's grid activation and factor's activation. Given the arrangement of vertex of graph G built with sampling and using (4), the characteritic polynomial of any of the vertex s composing layer 0 of G that correspond to bits sampled write as

$$\mathcal{P}_s = \mathcal{P}_{\{b_1^*, \dots, b_i^*, s\}, i+1}$$

As a consequence, the score process of each vertex sampled s in G is equal to the score process of a vertex $v(\{b_1^*, \dots b_i^*, s\}, i+1)$ at layer 1 of a graph of two layers where $\{b_1^*, \dots b_i^*, s\}$ composes its layer 0. Then, the first inequality is trivial, since it consists in a naive development. Indeed

$$\begin{aligned}\mathbb{P}(E) &= \sum_{j=0}^{|S|} p^j \times q^{|S|-j} \times \mathbb{P}_{\mathcal{S}_i}\left(|\{s \in S \setminus \omega_{I\cup\{s\},i+1,f} < \omega\}| = j\right) \\ &\leq \sum_{j=0}^{|S|} p^j \times \mathbb{P}_{\mathcal{S}}\left(|\{s \in S \setminus \omega_{I\cup\{s\},i+1,f} < \omega\}| = j\right)\end{aligned}$$

Where

- $I = \{b_1^*, \ldots, b_i^*\}$
- $p = \mathbb{P}\left(s_{v,f}(N,T,\rho) < 0 | \omega_{I\cup\{s\},i+1,f} < \omega\right)$
- $q = \mathbb{P}\left(s_{v,f}(N,T,\rho) < 0 | \omega_{I\cup\{s\},i+1,f} \geq \omega\right)$

Then, we choose the value of the postive real ω such that there exists a bit b^+ that verifies

$$\begin{aligned}b^+ = &\arg\min_{b\in\mathcal{G}} |\omega - \omega_{I\cup\{b\},i+1,f}| \\ &\textit{such that } \omega - \omega_{I\cup\{b\},i+1,f} > 0\end{aligned}$$

And we define the vertex $v^+(I \cup \{b^+\}, i+1)$ and $\delta^+ = |\omega - \omega_{v^+,f}|$. If vertex v is such that $\omega_{v,f} < \omega$ then using (5) one get

$$\underbrace{\frac{\|\mathcal{P}_f\|_{d'}}{\|\mathcal{P}_f\|_{d'} + (\omega - \delta) \times (1 - \|\mathcal{P}_f\|_{d'})}}_{\phi_{v,f}} \geq \underbrace{\frac{\|\mathcal{P}_f\|_{d'}}{\|\mathcal{P}_f\|_{d'} + (\omega - \delta^+) \times (1 - \|\mathcal{P}_f\|_{d'})}}_{\phi_{v^+,f}} \geq \underbrace{\frac{\|\mathcal{P}_f\|_{d'}}{\|\mathcal{P}_f\|_{d'} + \omega \times (1 - \|\mathcal{P}_f\|_{d'})}}_{\phi}$$

For some real $\delta > \delta^+ > 0$. Then, We choose the 3-tuple (N, T, ρ) as follow:

$$\begin{aligned}\rho &= \textit{min } \rho \in \mathbb{N} \textit{ such that } \phi \times (\rho + 1) - \rho < 0 \\ N &= -t \times (\phi \times (\rho + 1) - \rho) \\ T &= t\end{aligned}$$

for any $t \in \mathbb{N}^*$. Now considering v with $\omega_{v,f} < \omega$ one can write

$$\begin{aligned}\mathbb{P}\left(s_{v,f}[N,T,\rho] < 0\right) &= \mathbb{P}\left(N_i + \sum_{t=1}^{T} s_{v,\rho,T,f} < 0\right) \\ &= \mathbb{P}\left(\sum_{t=1}^{T} s_{v,\rho,t,f} - T \times \mathbb{E}\left[s_{v,\rho,1,f}\right] < -N - T \times \mathbb{E}\left[s_{v,\rho,1,f}\right]\right)\end{aligned}$$

Yet using equation (11) one have

$$\begin{aligned}\mathbb{E}\left[s_{v,\rho,1,f}\right] &= \phi_{v,f} \times (\rho+1) - \rho_i \\ &= \frac{\|\mathcal{P}_f\|_{d'}}{\|\mathcal{P}_f\|_{d'} + (\omega-\delta) \times (1-\|\mathcal{P}_f\|_{d'})} \times (\rho+1) - \rho\end{aligned}$$

Furthermore using $T = t$ and $N = t \times (\phi(\rho+1) - \rho)$ and the definition of ϕ one have

$$-N - T \times \mathbb{E}\left[s_{v,\rho,1,f}\right] = -t \times \delta \times \underbrace{(\rho+1) \times \phi \times \phi_{v,f} \times \frac{1-\|\mathcal{P}_f\|_{d'}}{\|\mathcal{P}_f\|_{d'}}}_{\tilde{\delta}_{i,f}}$$

Thus

$$\begin{aligned}\mathbb{P}\left(s_{v,f}[N,T,\rho] < 0\right) &= \mathbb{P}\left(\sum_{t=0}^{T-1} s_{v,\rho,t,f} - \mathbb{E}\left[s_{v,\rho,t,f}\right] < -t \times \delta \times \tilde{\delta_{i,f}}\right) \\ &\leq \mathbb{P}\left(|\sum_{t=0}^{T-1} s_{v,\rho,t,f} - \mathbb{E}\left[s_{v,\rho,t,f}\right]| > t \times \delta \times \tilde{\delta_{i,f}}\right) \\ &\leq \mathbb{P}\left(|\sum_{t=0}^{T-1} s_{v,\rho,t,f} - \mathbb{E}\left[s_{v,\rho,t,f}\right]| > t \times \delta^{+} \times \delta_{i,f}\right)\end{aligned}$$

With $\delta_{i,f} = (\rho_i + 1) \times \phi_i^2 \times \frac{1-\|\mathcal{P}_f\|_{d'}}{\|\mathcal{P}_f\|_{d'}}$.

At this point we have to notice that $\{s_{v,\rho,t,f}\}_{t=1,\ldots,T-1}$ is a sequence of i.i.d random variables with mean μ and variance σ^2 that verifies $|s_{v,\rho,t,f}| \leq \rho$. Thus one can apply the Chernoff inequality as formulated in [1]. In particular, taking $\lambda = \sigma^{-1}\delta^{+}\delta_{i,f}$ we obtain

$$\begin{aligned}\mathbb{P}\left(|\sum_{t=0}^{T-1} s_{v,\rho,tnf} - \mathbb{E}\left[s_{v,\rho,t,f}\right]| > t\delta^{+}\delta_{i,f}\right) &= \mathbb{P}\left(|\sum_{t=0}^{T-1} s_{v,\rho,t} - \mathbb{E}\left[s_{v,\rho,t}\right]| > \lambda\sigma\sqrt{t}\right) \\ &\leq C \times \max\left(\exp\left(-t \times \left(\frac{\delta^{+}\delta_{i,f}c}{\sigma}\right)^2\right), \exp\left(-t \times \delta^{+}\delta_{i,f}c\right)\right)\end{aligned}$$

With C, c some positive constant and $\mathrm{Var}[s_{v,\rho,t,f}] = \sigma^2$. Q.E.D.

Theorem 3.2 *Given a set of sampled bits S, a set of pre-selected bits $I = \{b_1^*, \ldots, b_i^*\}$ a target factor f and G the firing graph built after sampling algorithm with parameter N. There exists a sequence of 4-tuple (ω, N, T, ρ) such that for each remaining input vertex v of the drained firing graph verifies*

$$\mathbb{P}\left(\omega_{v,f} > \omega\right) \leq C \times \max\left(\exp\left(-t \times \left(\frac{\delta^{-}\delta_{i,f}c}{\sigma}\right)^2\right), \exp\left(-t \times \delta^{-}\delta_{i,f}c\right)\right) \tag{13}$$

With $\delta_{i,f}$, δ^{+} C and c are postitive constants that depends on ω and i and $\mathrm{Var}[s_{v,\rho,t}] = \sigma^2$.

Proof. First recalling the topology of the firing graph G built at sampling step, again, using a similar argument (4) than in the previous proof, for each sampled bits $s \in S$ and each input vertex $v(\{s\}, 1)$ of G $\omega_{v,f} = \omega_{I\cup\{s\},i+1,f}$. Furthermore

$$\mathbb{P}\left(\omega_{v,f} > \omega\right) = \mathbb{P}_{\mathcal{S}}\left(\omega_{I\cup\{s\},i+1,f} < \omega\right) \times \mathbb{P}\left(s_{v,f}(N,T,\rho) > 0 | \omega_{I\cup\{s\},i+1,f} > \omega\right)$$

Where $s \in S$ and $v(I \cup \{s\}, i+1)$ a vertex of layer 1 of a firing graph with layer 0 composed of $I \cup \{s\}$. Then, we choose the value of the postive real ω such that there exists a bit b^- that verifies

$$\begin{aligned} b^- &= \arg\min_{b\in\mathcal{G}} |\omega - \omega_{I\cup\{b\},f}| \\ &\quad \textit{such that } \omega - \omega_{I\cup\{b\},f} < 0 \end{aligned}$$

And we define the vertex $v^-(I \cup \{b^-\}, i+1)$ and $\delta^- = |\omega - \omega_{v^-,f}|$. If v is such that $\omega_{v,f} < \omega$ then using (5) we have

$$\underbrace{\frac{\|\mathcal{P}_f\|_{d'}}{\|\mathcal{P}_f\|_{d'} + (\omega+\delta)\times(1-\|\mathcal{P}_f\|_{d'})}}_{\phi_{v,f}} \leq \underbrace{\frac{\|\mathcal{P}_f\|_{d'}}{\|\mathcal{P}_f\|_{d'} + (\omega+\delta^-)\times(1-\|\mathcal{P}_f\|_{d'})}}_{\phi_{v^-,f}} \leq \underbrace{\frac{\|\mathcal{P}_f\|_{d'}}{\|\mathcal{P}_f\|_{d'} + \omega\times(1-\|\mathcal{P}_f\|_{d'})}}_{\phi}$$

for some $\delta \geq \delta^- > 0$. Then defining the 3-tuple (N, T, ρ) as

$$\begin{aligned} \rho &= \textit{min } \rho \in \mathbb{N} \textit{ such that } \phi \times (\rho+1) - \rho < 0 \\ N &= -t \times (\phi \times (\rho+1) - \rho) \times \mu_{v,f} \\ T &= t \end{aligned}$$

for any $t \in \mathbb{N}^*$. Now considering v with $\omega_{v,f} > \omega$, reproducing the same development as it was done in the proof of previous Theorem, one can derive a convenient form to easily apply the Chenoff inequality.

$$\mathbb{P}\left(s_{v,f}(N,T,\rho_i) > 0 | \omega_{v,f} > \omega\right) \leq \mathbb{P}\left(|\sum_{t=0}^{T_i-1} s_{v,\rho,t,f} - \mathbb{E}\left[s_{v,\rho,t,f}\right]| > t \times \delta^- \delta_{i,f}\right)$$

With $\delta_{i,f} = (\rho+1) \times \phi^2 \times \frac{1-\|\mathcal{P}_f\|_{d'}}{\|\mathcal{P}_f\|_{d'}}$. Then using the Chernoff inequality as written in [1] using $\lambda = \sigma^{-1}\delta^+\delta_{i,f}$ we obtain

$$\mathbb{P}\left(|\sum_{t=0}^{T-1} s_{v,\rho,t,f} - \mathbb{E}\left[s_{v,\rho,t,f}\right]| > T \times \mu_{v,f}\delta^-\rho\right) \leq C \times \max\left(\exp\left(-t \times \left(\frac{\delta^-\delta_{i,f}c}{\sigma}\right)^2\right), \exp\left(-t \times \delta^-\delta_{i,f}c\right)\right)$$

With C, c some positive constant and $\mathrm{Var}[s_{v,\rho,t,f}] = \sigma^2$. Q.E.D

3.4 Limit of the generic case

The combination of Theorems shows that, the association of sampling and draining with the right choice of 4-tuple (ω, N, T, ρ) gives a convenient tool to identify measure grid'sbit with purity coefficient larger or equal to ω. Furthermore when $t \to +\infty$ the identification is almost certain, which highlights the trade-off between efficiency and complexity of the algorithm embedded in the choice of ω and t, on which depends N and T. This generic case and its analysis deliver a strong framework that ease the derivation of more specific result that may be obtained under specific distribution of latent factors activation and link to measure grid. Nevertheless it leaves two fundamental points clueless

- No possibility to quantify further the effectivenes of the sampling strategy
- Is there any good procedure or heuristic to choose positive real value ω ?

In the rest of this paper, we present two different specific cases of factor's and measure grid's modelisation that enable a more specific of both the quantification of the sampling strategy and choice of ω.

4 Case of signal plus noise

This particular case is designed to be easy to analyze. We first define the statistical signature of factor and bit's activation. Then we quantify the sampling strategy and args a choice for the sequence 4-tuple (ω, T, N, ρ). Finally we present simulations and provide discussion of the result obtained with this special case.

4.1 Statistical modelisation

In this particular case we assume that the target factor f is linked to some $|\mathcal{G}(f)|= k$ measure grid's bits and activate with probability p_f. We also assume that bits of the measure grid are identically and independently subject to a noisy activation with probability p_N. We may see noisy activations as the result of activation of n noisy latent factor which are linked to exactly 1 bit of the measure grid, that is $K = n + 1$. Under this modelisation, the probability for a bit $b \in \mathcal{G}$ to activate is defined as

$$\mathbb{P}\left(\text{"}b \textit{ active}\text{"}\right) = \begin{cases} p_f + p_N \times (1 - p_f), & \text{if } \ b \in \mathcal{G}(f^*) \\ p_N, & \text{Otherwise} \end{cases}$$

As a consequence, for any $I \in S(\mathcal{G})$ such that $|I \cap \mathcal{G}(f)|= i$ and $j = |I|-i$ if we set $x \in F(2)^n$ such that $x \in F_2(\{I\}, \mathcal{G})$, the distribution over measure grid bit's activation is defined as

$$d_x = \begin{cases} p_N^{i+j} \times (1 - p_N)^{n-i-j} \times (1 - p_*) + p_N^{j} \times (1 - p_N)^{n-k-j} \times p_*, & \text{if } \ i = k \\ p_N^{i+j} \times (1 - p_N)^{n-i-j} \times (1 - p_*), & \text{Otherwise} \end{cases}$$

The rest of the section will always refer to this distribution as d.

4.2 Evaluation of bits

Let G be a firing graph with a layer 0 composed of measure grid's bits. Then, the precision of a vertex $v(I, |I|)$ of layer 1 of G with respect to f depends only on $|I|$ and $|I \cap \mathcal{G}(f)|$. Indeed, if $|I \cap \mathcal{G}(f)| = i$ then

$$\phi_{v,f} = \frac{p_f}{p_f + (1 - p_f) \times p_N^i}$$

With identification of terms using (5) we have $\omega_{v,f} = p_N^i$ and using previously defined distribution, one find that $\mu_{v,f} = p_N^{|I|-i}$, $\nu_{v,f} = p_N^{|I|}$. Besides, given a set of bits I, such that $I \subset \mathcal{G}(f)$, for any bit $b \in \mathcal{G} \setminus I$ if $b \in \mathcal{G}(f)$ the precision of vertex $v(I \cup \{b\}, |I|+1)$ with respect to f is

$$\phi_v = \frac{p_f}{p_f + (1 - p_f) \times p_N^{|I|+1}}$$

if $b \notin \mathcal{G}(f)$

$$\phi_v = \frac{p_f}{p_f + (1 - p_f) \times p_N^{|I|}}$$

4.3 Sampling Strategy

In this particular case we follow the generic sampling procedure $\mathcal{S}$ with parameter $p_{\mathcal{S}}$. Thus, using the previously defined statistical distribution of bits activation, if we denote S the set of sampled bits using $\mathcal{S}$, then the distribution of the cardinal of S is

$$\mathbb{P}(|S| = s) = \begin{cases} \binom{n-k}{s-k} \times p_N^{s-k} \times (1 - p_N)^{n-k-s} \times p_{\mathcal{S}}, & \text{if } s \geq k \\ 0, & \text{otherwise} \end{cases}$$

Thus its expected size is $\mathbb{E}[|S|] = k + (n - k) \times p_N \times p_{\mathcal{S}}$. Furthermore if $I = \{b_1^*, \ldots, b_i^*\} \in S(\{)$ a set of pre-selected bits and S is a set of bits sampled using $\mathcal{S}$, then, there exists a positive real ω_i such that

$$\begin{aligned} \mathbb{P}_{\mathcal{S}}\left(|\{s \in S \setminus \omega_{I \cup \{s\}, |I|+1, f} < \omega_i\}| = j\right) &= \mathbb{P}_{\mathcal{S}}\left(|\{s \in S \setminus s \in \mathcal{G}(f)| = j\right) \\ &= \binom{|\mathcal{G}(f)|-i}{j} \times p_{\mathcal{S}}^j \times (1 - p_{\mathcal{S}})^{|\mathcal{G}(f)|-i-j} \end{aligned}$$

4.4 Identification of factors

First, in the case of a single sampled firing w, one can see that bit's purity coefficients can take only two values with respect to f

$$\omega_{\{b\},1,f} = \begin{cases} p_N, & \text{if } b \in \mathcal{G}(f) \\ 1, & \text{otherwise} \end{cases}$$

Thus if we choose

$$\omega_0 = \frac{(1+p_N)}{2}$$

It maximizes the purity margin defined as

$$\delta = \frac{(\omega_0 - \omega_{\{b\},1,f}) + (\omega_{\{b'\},1,f} - \omega_0)}{2} = \frac{(1-p_N)}{2}$$

Where $b \in \mathcal{G}(f)$ and $b' \notin \mathcal{G}(f)$. In the case of a joint sampled firing graph in which a set $I = \{b_1^*, \ldots, b_i^*\}$ of $i \in \mathbb{N}^*$ pre selected bits that verifies $\forall b \in I$, $b \in \mathcal{G}(f)$. Then remaining bit's purity coefficients with respect to f can take again two values

$$\omega_{I\cup\{b\},i+1,f} = \begin{cases} p_N^{i+1}, & \text{if } \ b \in \mathcal{G}(f) \\ p_N^{i}, & \text{otherwise} \end{cases}$$

Thus if we choose

$$\omega_i = \frac{(1+p_N) \times p_N^i}{2}$$

it maximizes the purity margin defined as

$$\delta = \frac{(\omega_i - \omega_{I\cup\{b\},i+1,f}) + (\omega_{I\cup\{b'\},i+1,f} - \omega_i)}{2} = \frac{(1-p_N) \times p_N^i}{2}$$

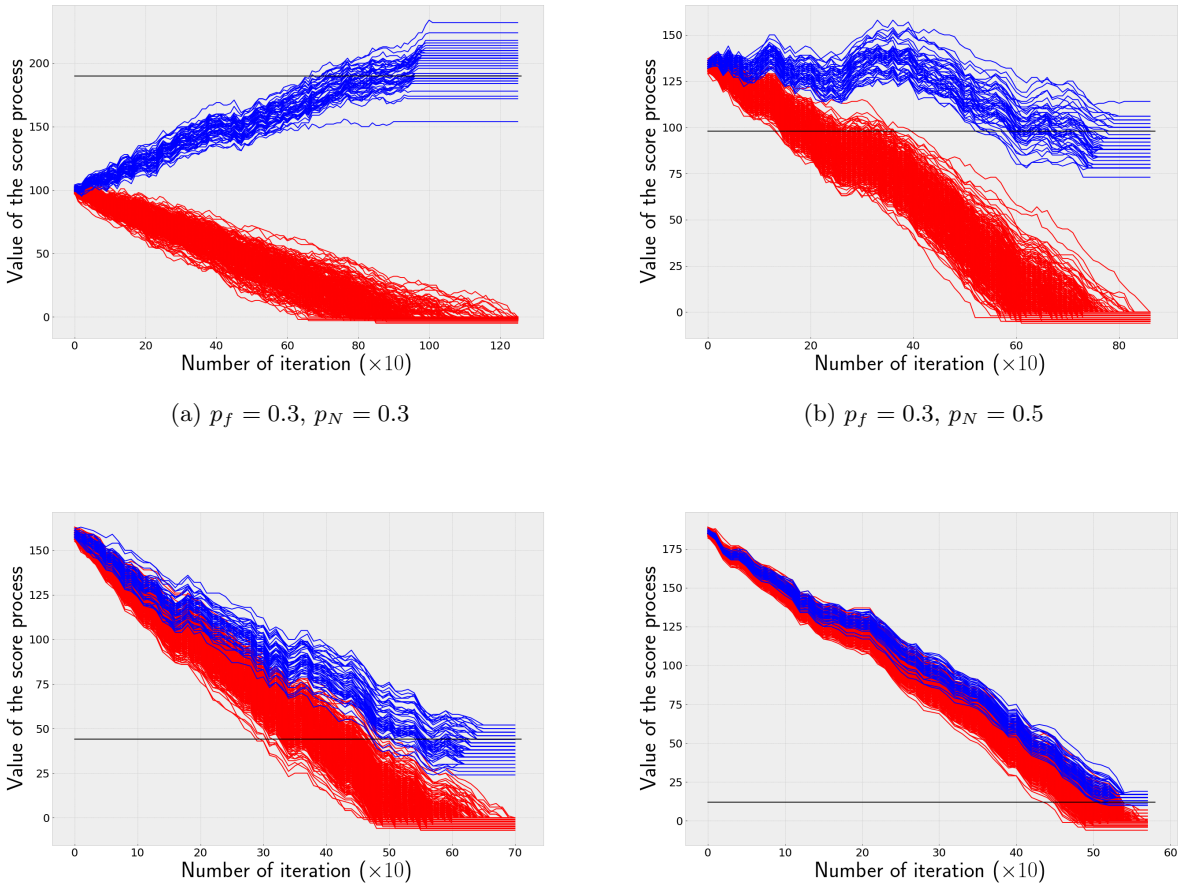
Where $b \in \mathcal{G}(f)$ and $b \notin \mathcal{G}(f)$. Finally we define the 4-tuple (N_i, T_i, ρ_i) as

$$\begin{aligned} \rho_i &= min\ \rho \in \mathbb{N}\ such\ that\ \phi_i \times (\rho+1) - \rho < 0 \\ N_i &= -t \times (\phi_i \times (\rho_i + 1) - \rho_i) \\ T_i &= t \end{aligned}$$

Where $t \in \mathbb{N}$ and $\phi_i = \frac{p_f}{p_f + \omega_i \times (1-p_f)}$.

4.5 Simulation

The signal plus noise model is implemented in python and mainly use standard numpy module to generate random signal that fit its probabilistic framework, more details about the implementation can be found in appendix B. The simulation of this signal plus noise requires the implementation of random signals that respect probabilistic properties of the model. In our case we generate $n = 1000$ bits that randomly activate with probbility p_N and we choose randomly $|\mathcal{G}(f)|= 50$ bits that are linked to a latent factor that activate with probability $p_f = 0.3$. Finally we built the single sampled firing graph running the sampling algorithm described previously using $p_{\mathcal{S}} = 1$.



(a) $p_f = 0.3$, $p_N = 0.3$

(b) $p_f = 0.3$, $p_N = 0.5$

(c) $p_f = 0.3$, $p_N = 0.7$

(d) $p_f = 0.3$, $p_N = 0.9$

Figure 6: Observation of the score process for different SNR models $T = 500$

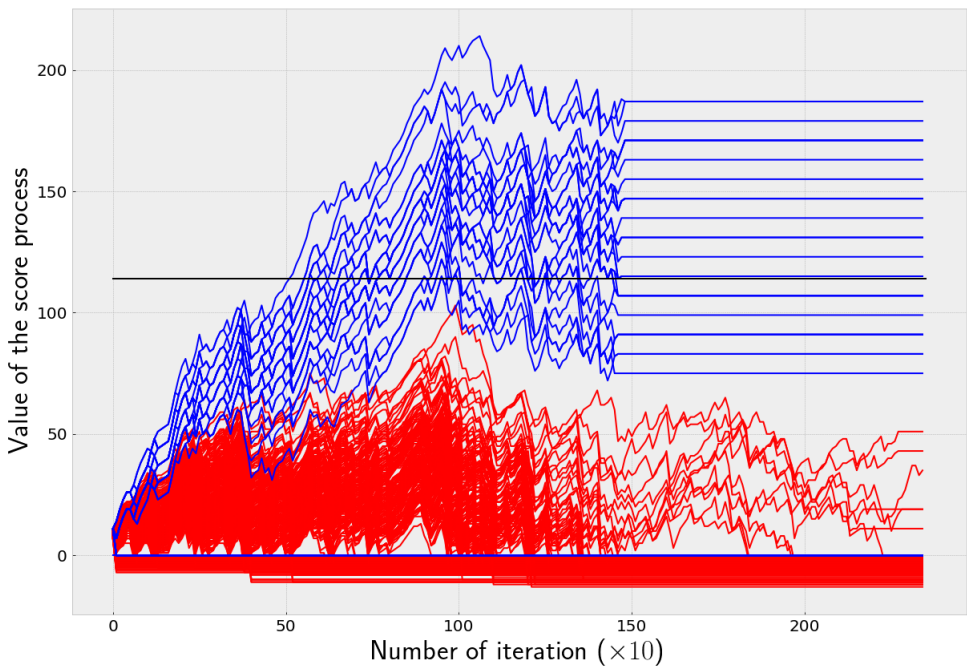
In every experiement, following the procedure described in the previous section we end with $\rho = 1$. Furthermore, each of the figure above shows the values of the score process through the tracking of the weight of edges of a single sampled firing graph. Blue lines show the weight of edges outcoming from input vertices whose activation corresponds to activation of some bits $b \in \mathcal{G}(f)$. Red lines correspond to the weight of edges outcoming from input vertices whose activation corresponds to noisy activation. Finally the black horizontal line represents the theoretical mean value of of the score process, with respect to f, with parameter N, T, ρ chosen as described previously, of an edge outcoming from a input vertex that activates as a bits of the grids linked to f. As theoretical modelisation suggests we have a clear separation between the two kind of bits and a convergence toward theoretical mean for process corresponding to factors activation. Yet the higher is the noisy activation probability the more difficult it is to separate the tow category of bits sampled. In order to highlight it we run a new simulation with $n = 1000$, $|\mathcal{G}(f)|= 50$, $p_f = 0.3$, $T = 200$ and $p_{\mathcal{S}}$. Yet at the end of the draining we choose all the input vertices of the firing graph that stil have outcoming edges, and use their combined activation as estimator of the factor's activation measuring their precision and recall over 100 repetion for each SNR ratio

Table 1: Evaluation of naive factor's activation estimation, $T = 200$, 100 repetitions

P_N	Mean ϕ	Standard deviation ϕ	Mean ψ	Standard deviation ψ	Number of fails
0.3	1.0	0.0	0.7	0.29	0
0.5	1.0	0.03	0.6	0.42	0
0.7	0.97	0.1	0.46	0.46	4
0.9	0.75	0.15	0.32	0.32	19

The table above show quality indicator of the estimator for different SNR ratio. the two first columns gives respectively the mean and standard deviation of the precision of the estimator. The two following columns are respectively the mean and the standard deviation of the recall of the estimator. Finally the last column is the number of experiments that ended without any remaining input vertex, so that the construction of an estimator is not possible. Again, we see that the quality of the estimator drops as the theoretical precision between noisy bits and factor's bits are close to each other. Yet it reveal that this naive estimator, for a reasonable SNR ratio, is still efficient to predict the activation of target latent factor. Finally we run the same experiment in the settings of joint sampled firing graph using again a measure grid of $n = 1000$ bits from which we sampled randomly $|\mathcal{G}(f)|= 50$ bits linked to to target factor f that activate with probability $p_f = 0.3$ and we set $p_N = 0.6$. Finally we built the joint sampled firing graph by pre-selecting randomly 5 bits linked to the factor and running the sampling algorithm described previously using $p_{\mathcal{S}} = 1$.

Figure 7: Observation of the score process in a joint sampled firing graph with $T = 500$



In this case, following the procedure described previously we end with $N = 7$, $\rho = 7$ and $\omega_5 \simeq 0.062$, those value of ρ the gain in variance of score process that we observe. In this case we are more likely to drain vertex that activate corrrectly with factors and we are more likely to have noisy vertex remaining after the draining.

5 Case of sparse measure grid

This particular case is a bit more complex since we assume randomness of linked between latent factor and measure grid's bits. Again, for this particular case, we first define the statistical signature of factor and bit's activation. Then we quantify the sampling strategy and args a choice for the sequence 4-tuple (ω, T, N, ρ). Finally we present simulations and provide discussion of the result obtained with this sparse case.

5.1 Statistical modelisation

Latent factor activation

We assume that activation of each of the K latent factors is identical and independant. We denote by p_f the probability of a latent factor's activation. As a consequence, for any $I \in S(\mathcal{F})$, if we define x such that $x \in F_2(\{I\}, \mathcal{F})$ then we can define the distribution of factor's activation as

$$d'_x = \times p_f^{|I|} \times (1 - p_f)^{K-|I|}$$

Measure grid activation

We assume two major properties of activation of measure grid's bits.

- For each factor $f \in \mathcal{F}$, each bit $b \in \mathcal{G}$ has equal probability p_g to belong to $\mathcal{G}(f)$.
- For each factor $f \in \mathcal{F}$, for each couple $b_1, b_2 \in \mathcal{G}^2$, events "$b_1 \in \mathcal{G}(f)$" and "$b_2 \in \mathcal{G}(f)$" are independent.

As a consequence the probability for a bit b to activate, given that every factor of some set $\{f_1, \ldots, f_k\} \subset \mathcal{F}$ is active write

$$\begin{aligned}\mathbb{P}\left(b \textit{ active } | f_1, \ldots, f_k \textit{ active}\right) &= \sum_{i=1}^{k} \binom{k}{i} \times p_g^i \times (1 - p_g)^{k-i} \\ &= 1 - (1 - p_g)^k\end{aligned}$$

The above quantity depends only on the number of active latent factor. Thus, for any $I \in S(\mathcal{G})$, if we define x such that $x \in F_2(S(\mathcal{G}), \mathcal{G})$ then we can define the distribution of bits's activation as

$$d_x = \sum_{k=1}^{K} \left[\binom{K}{k} \times p_f^k \times (1-p_f)^{K-k} \right] \times \left[p_{g|k}^i \times (1-p_{g|k})^{n-i} \right]$$

With $i = |I|$ and $p_{g|k} = \mathbb{P}\left(b \ active \ |f_1, \ldots, f_k \ active\right)$.

5.2 Evaluation of bits

Let G be a firing graph whose layer 0 is composed of measure grid's bits, given a target factor f, the precision with respect to f of a vertex $v(\{b\}, 1)$ of the layer 1 of G depends on $b \in \mathcal{G}(f)$ and $|\mathcal{G}^{-1}(b)|$. Indeed if $b \in \mathcal{G}(f)$ and $|\mathcal{G}^{-1}(b)|= l$, $l \in \{1, \ldots, K\}$, it will be said to have a purity rank of l and its precision with respect to f writes

$$\phi_{v,f} = \frac{p_f}{p_f + (1-p_f) \times \omega_{v,f}}$$

where $\omega_l = 1 - (1-p_f)^{l-1}$. If $b' \notin \mathcal{G}(f)$ and $|\mathcal{G}^{-1}(b')|= l$ then $v'(\{b'\}, 1)$ verifies $\mu_{v',f} = \nu_{v',f} = (1-(1-p_f)^l)$. Thus its precision with respect to f writes

$$\phi_{v',f} = p_f$$

Futhermore if we have a vertex $v(I, |I|)$ such that $\forall b \in I$, $b \in \mathcal{G}(f)$ and $\min_{b \in I} |\mathcal{G}^{-1}(b)|= l$, then the precision of v with respect to f verifies

$$\phi_{v,f} \leq \frac{p_f}{p_f + \omega_l^- \times (1-pf)}$$

With

$$\omega_l^- = \sum_{k=K-l-1}^{K} \binom{K}{k} p_f^k \times (1-p_f)^{K-k}$$

The minimum possible purity coefficient one can obtained with bits that verifies $b \in \mathcal{G}(f)$ and $|\mathcal{G}^{-1}(b)|= l$. That is the case of a vertex $v(I, |I|)$ with I composed of every possible $\binom{K}{l}$ such bits.

5.3 Sampling Strategy

Again here we follow the generic sampling procedure $\mathcal{S}$ with parameter $p_{\mathcal{S}}$. Although it is not hard to derive key quantification such as $\mathbb{E}\left[|S|\right]$ or probabilities to sample bits linked to a target factor f under this modelisation, the generic formulas are not elegant and present not much interest in this simulation.

5.4 Identification of factors

First, in the case of a single sampled firing, for any grid's bits linked to factor f can have K different purity coefficients related to their purity rank described previously. Thus we may set ω_0 to ω_l, using l reasonably small so to differentiate lower or similar purity rank from greater

purity rank samples. In the case of a joint sampled graph, where a set of $I = \{b_1^*, \dots b_i^*\}$ were pre-selected then the choice of ω_i is not trivial and cannot be efficiently and generically derived. Let $\omega_{I,|I|,f}$ the purity coefficient of the preselected set of bits we set $\omega_i = \omega_{I,|I|,f} + \delta$ with smaller δ with larger i, so that the draining discriminate between set of bits whose joint purity coefficient is above.vFinally we choose the sequence of 4-tuple $\{(\omega_i, N_i, T_i, \rho_i)\}_{i=1,\dots}$ as

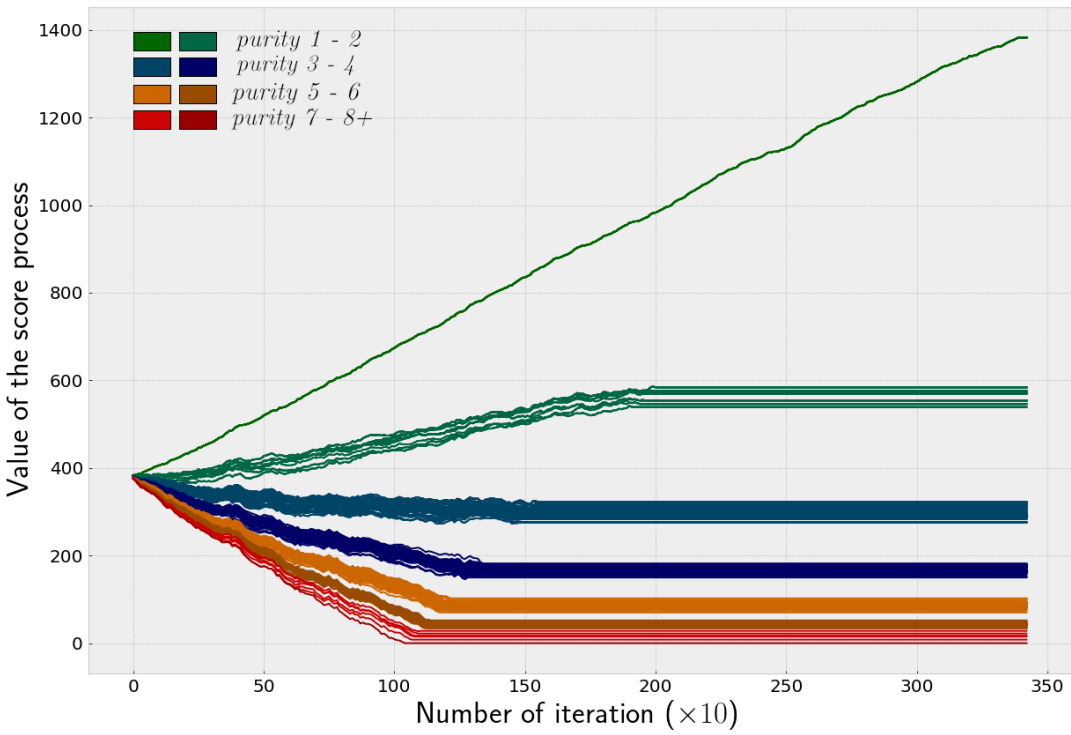
$$\begin{aligned}
\rho_i &= min\ \rho \in \mathbb{N}\ such\ that\ \phi_i \times (\rho + 1) - \rho < 0 \\
N_i &= -t \times (\phi_i \times (\rho_i + 1) - \rho_i) \\
T_i &= t
\end{aligned}$$

For some $t \in \mathbb{N}$ and $\phi_i = \frac{p_f}{p_f + \omega_i \times (1 - p_f)}$.

5.5 Simulation

Here again, the sparse measure grid model is implemented using python and the standard numpy.random module to generate random signal that fit its probabilistic framework. In our case we generate $n = 1000$ bits. Then for each of the $K = 10$ latent factor we link measure grid's bits independently with probabiity $p_g = 0.3$. Furthermore each latent factor has the same probability $p_f = 0.3$ to activate. Finally we built the single sampled firing graph running the sampling algorithm described previously using $p_{\mathcal{S}} = 1$ and we set $\omega = \omega_{10}$ the lowest purity coefficient for a bits that is linked to the target factor.

Figure 8: Observation of the score process in a sing sampled firing graph with $T = 1000$



We clearly see a rapid differentiation of score process according to their purity rank. We can also observe that the lower the coefficient the closer are the score process of corresponding edges. Finally the behaviour of score processus validate the efficiency of the draining algorithm to rank bits of the measure grids blindly in a attempt to identify latent factor.

MAY BE ADD A FUCKING JOINT SAMPLED FIRING GRAPH DRAINING

6 Discussion and future work

A Properties

<u>Partition</u>

$v_1(I, l_0)$, $v_2(J, 0)$ and $v_3(K, 0)$ are three vertices of the same firing graph, with the same input domain $\mathcal{G}$ such that $I = J \cup K$ and $J \cap K = \emptyset$. result (1) stands that $\forall x \in F_2(S(\mathcal{G}), \mathcal{G})$

$$\mathcal{P}_I^{l_0}[x] = \sum_{l=l_0}^{|I|} \sum_{j=0}^{|J|} \mathcal{P}_{J,j}[x] \cdot \mathcal{P}_{K,l-j}[x]$$

Proof. The statement above can also be written

$$\mathcal{P}_{I,l}[x] = \sum_{j=0}^{|J|} \mathcal{P}_{J,j}[x] \cdot \mathcal{P}_{K,l-j}[x]$$

$\forall\, l \in \{l_0, \ldots, |I|\}$, now we propose a simple proof by contradiction, there exist a $l \in \{l_O, \ldots, |I|\}$ such that there exist $X \in S(\mathcal{G})$, $x \in F_2(\{X\}, \mathcal{G})$ and

$$\mathcal{P}_{I,l}[x] = 1$$
$$\sum_{j=0}^{|J|} \mathcal{P}_{J,j}[x] \cdot \mathcal{P}_{K,l-j}[x] = 0$$

J and K compose a partition of I and $|I \cap X| = l$, thus there exists a couple (j^*, k^*) such taht

$$|X \cap J| = j^*$$
$$|X \cap K| = k^*$$
$$j^* + k^* = l$$

Thus for $x \in F_2(\{X\}, \mathcal{G})$

$$\mathcal{P}_{J,j^*}[x] \cdot \mathcal{P}_{K,k^*}[x] = 1$$

which contradict our first assumption. Now assume there exist a $l \in \{l_O, \ldots, |I|\}$ such that there exist $X \in S(\mathcal{G})$, $x \in F_2(\{X\}, \mathcal{G})$

$$\mathcal{P}_{I,l}[x] = 0$$
$$\sum_{j=0}^{|J|} \mathcal{P}_{J,j}[x] \cdot \mathcal{P}_{K,l-j}[x] = 1$$

Thus above statement implies that there exists a $j^* \in \{0, \ldots, min(l, |J|)\}$ such that

$$\mathcal{P}_{J,j^*}[x] \cdot \mathcal{P}_{K,l-*}[x] = 1$$

Thus $|X \cap J| = j^*$ and $|X \cap K| = l - j^*$. Since J and K is a partion of I we must have

$$|X \cap I| = |X \cap J| + |X \cap K| = l$$

As a consequence for $x \in F_2(\{X\}, \mathcal{G})$, $\mathcal{P}_{I,l}[x] = 1$ and give us the contradiction.

Result (2) is a particular case of result (1)

<u>Decomposition</u>

G is a firing graph composed of a set of $\mathcal{G}$ of vertice of layer 0. $u(I, l_u)$, $v(I', l_v)$ such that $I \cap I' = \emptyset$ as vertice of layer 1 and $w(\{u, v\}, 2)$ as vertex of layer 2. Let $K \in \cup_{l \in \{l_v, \ldots, |I'|\}} S(I', l)$, $x \in F_2(S(\mathcal{G}), \mathcal{G})$ and $x' = \begin{bmatrix} \mathcal{P}_u[x] & \mathcal{P}_v[x] \end{bmatrix}$, the result (3) stands

$$\mathcal{P}_{K,|K|}[x] \cdot \mathcal{P}_{\{u,v\},2}[x'] = \sum_{l=l_u}^{|I|} \sum_{J \in S(I,l)} \mathcal{P}_{J \cup K, l+|K|}[x]$$

Proof. The proof the above statement is derived by a straight forward development of the equation, first using result (3) and the fact that K and $I' \setminus K$ is a partition of I' we can write

$$\begin{aligned}
\mathcal{P}_{\{u,v\},2}[x'] &= \left(\sum_{l=l_u}^{|I|} \mathcal{P}_{I,l}[x] \right) \cdot \left(\sum_{l=l_v}^{|I'|} \mathcal{P}_{I',l}[x] \right) \\
&= \left(\sum_{l=l_u}^{|I|} \mathcal{P}_{I,l}[x] \right) \cdot \left(\sum_{l=l_v}^{|I'|} \sum_{k=0}^{|K|} \mathcal{P}_{K,k}[x] \cdot \mathcal{P}_{I \setminus K, l-k}[x] \right)
\end{aligned}$$

Thus

$$
\begin{aligned}
\mathcal{P}_{K,|K|}\left[x\right]\cdot\mathcal{P}_{\{u,v\},2}\left[x'\right] &= \mathcal{P}_{K,|K|}\left[x\right]\cdot\left(\sum_{l=l_u}^{|I|}\mathcal{P}_{I,l}\left[x\right]\right)\cdot\left(\sum_{l=l_v}^{|I'|}\sum_{k=0}^{|K|}\mathcal{P}_{K,k}\left[x\right]\cdot\mathcal{P}_{I\setminus K,l-k}\left[x\right]\right)\\
&= \left(\sum_{l=l_u}^{|I|}\mathcal{P}_{I,l}\left[x\right]\right)\cdot\left(\sum_{l=l_v}^{|I'|}\mathcal{P}_{K,|K|}\left[x\right]\cdot\mathcal{P}_{I\setminus K,l-|K|}\left[x\right]\right)\\
&= \left(\sum_{l=l_u}^{|I|}\mathcal{P}_{I,l}\left[x\right]\cdot\mathcal{P}_{K,|K|}\left[x\right]\right)\cdot\underbrace{\sum_{l=0}^{|I'|-|K|}\mathcal{P}_{I\setminus K,l}\left[x\right]}_{=1}\\
&= \sum_{l=l_u}^{|I|}\mathcal{P}_{I,l}\left[x\right]\cdot\mathcal{P}_{K,|K|}\left[x\right]
\end{aligned}
$$

The last line is equal to $\sum_{l=l_u}^{|I|}\sum_{J\in S(I,l)}\mathcal{P}_{J\cup K,l+|K|}\left[x\right]$ and thus the proof is achieved.

Result (3) is a particular case of result (4)

<u>Precision of vertex</u>

Let G be a firing graph with layer 0 composed by measure grid's bits $\mathcal{G}$ and $f\in\mathcal{F}$ denote some target factor that is linked to some bit of the measure grid. The distribution of activation of latent factors and measure grid's bits will be denoted d and d' and the event "factor f is active" will be denoted by e. Let v be a vertex connected to a set I of measure grid's bits with level $l\in\{1,\ldots,|I|\}$ and let $f\in\mathcal{F}$ be some factor. The result (5) stands that the precision of v with respect to f writes

$$\phi_{v,f}=\frac{\|\mathcal{P}_f\|_{d'}}{\|\mathcal{P}_f\|_{d'}+(1-\|\mathcal{P}_f\|_{d'})\times\omega_{v,f}}$$

Furthermore, the result (6) stands that if $\mu_{v,f}=1$ we have

$$\phi_{v,f}\leq\frac{\|\mathcal{P}_f\|_{d'}}{\|\mathcal{P}_f\|_{d'}+(1-\|\mathcal{P}_f\|_{d'})\times\omega_{\mathcal{G}(f),|\mathcal{G}(f)|,f}}$$

Proof. First, starting from the defintion of $\phi_{v,f}$

$$\phi_{v,f}=\phi_{I,l,f}=\frac{\|\mathcal{P}_I^l\|_{d,e}}{\|\mathcal{P}_I^l\|_d}$$

Thus using $\mathcal{P}_*=\mathcal{P}_{\mathcal{G}(f),|\mathcal{G}(f)|}$ one have

$$\phi_{v,f}=\frac{\langle\mathcal{P}_I^l,\mathcal{P}_*\rangle_{d,e}+\langle\mathcal{P}_I^l,\bar{\mathcal{P}}_*\rangle_{d,e}}{\langle\mathcal{P}_I^l,\mathcal{P}_*\rangle_{d,e}+\langle\mathcal{P}_I^l,\bar{\mathcal{P}}_*\rangle_{d,e}+\langle\mathcal{P}_I^l,\mathcal{P}_*\rangle_{d,\bar{e}}+\langle\mathcal{P}_I^l,\bar{\mathcal{P}}_*\rangle_{d,\bar{e}}}$$

Yet $\langle\mathcal{P}_I^l,\bar{\mathcal{P}}_*\rangle_{d,e}=0$

$$\phi_{v,f} = \frac{\langle \mathcal{P}_I^l, \mathcal{P}_* \rangle_{d|e} \times \|\mathcal{P}_f\|_{d'}}{\langle \mathcal{P}_I^l, \mathcal{P}_* \rangle_{d|e} \times \|\mathcal{P}_f\|_{d'} + \left(\langle \mathcal{P}_I^l, \mathcal{P}_* \rangle_{d|\bar{e}} + \langle \mathcal{P}_I^l, \bar{\mathcal{P}}_* \rangle_{d|\bar{e}} \right) \times \|\bar{\mathcal{P}}_f\|_{d'}}$$

Finally by identification of term

$$\phi_{v,f} = \frac{\mu_{I,l,f} \times \|\mathcal{P}_f\|_{d'}}{\mu_{I,l,f} \times \|\mathcal{P}_f\|_{d'} + \nu_{I,l,f} \times (1 - \|\mathcal{P}_f\|_{d'})}$$

With the same context, the result (6) stands that if $\mu_{v,f} = 1$ we have

$$\phi_{v,f} \leq \frac{\|\mathcal{P}_f\|_{d'}}{\|\mathcal{P}_f\|_{d'} + (1 - \|\mathcal{P}_f\|_{d'}) \times \omega_{\mathcal{G}(f),|\mathcal{G}(f)|,f}}$$

Proof. The result (6) can be proven by simple contradiction, suppose there is a tuple $(I, l_0) \neq (\mathcal{G}(f), |\mathcal{G}(f)|)$ such that

$$\mu_{I,l_0,f} = 1$$
$$\nu_{I,l_0,f} < \nu_{\mathcal{G}(f),|\mathcal{G}(f)|,f}$$

First, denoting $\mathcal{P}_{\mathcal{G}(f),|\mathcal{G}(f)|} = \mathcal{P}_*$ and using (1) we have

$$\begin{aligned}
\langle \mathcal{P}_I^{l_0}, \mathcal{P}_* \rangle_d &= \sum_{l=l_0}^{|I|} \sum_{k=0}^{I \cap \mathcal{G}(f)} \langle \mathcal{P}_{I \cap \mathcal{G}(f),k} \cdot \mathcal{P}_{I \setminus \mathcal{G}(f),l-k}, \mathcal{P}_* \rangle_d \\
&= \sum_{l=l_0}^{|I|} \sum_{k=0}^{I \cap \mathcal{G}(f)} \sum_{x \in F_2(S(\mathcal{G}),\mathcal{G})} \mathcal{P}_{I \cap \mathcal{G}(f),k}[x] \cdot \mathcal{P}_{I \setminus \mathcal{G}(f),l-k}[x] \cdot \mathcal{P}_*[x] \times d_x \\
&= \sum_{x \in F_2(S(\mathcal{G}),\mathcal{G})} \mathcal{P}_*[x] \cdot \left(\sum_{l=l_0}^{|I|} \mathcal{P}_{I \setminus \mathcal{G}(f),l-|I \cap \mathcal{G}(f)|}[x] \right) \times d_x
\end{aligned}$$

Where d is any well defined ditribution on $F_2(S(\mathcal{G}), \mathcal{G})$. Thus, we have

$$\mu_{I,l_0,f} = \sum_{x \in F_2(S(\mathcal{G}),\mathcal{G})} \mathcal{P}_*[x] \cdot \left(\sum_{l=l_0}^{|I|} \mathcal{P}_{I \setminus \mathcal{G}(f),l-|I \cap \mathcal{G}(f)|}[x] \right) \times d_{x|e}$$

As a consequence, in order to have $\mu_{I,l_0,f} = 1$ we most have $\{\mathcal{P}_{I \setminus \mathcal{G}(f),l-|I \cap \mathcal{G}(f)|}\}_{l \in \{l_0,\ldots,|I|\}}$ to be a partion of $F_2(S(\mathcal{G}), \mathcal{G})$. Thus $l_0 \leq |I \cap \mathcal{G}(f)|$. On the other end the precision coefficient writes

$$\nu_{I,l_0,f} = \sum_{x \in F_2(S(\mathcal{G}),\mathcal{G})} \mathcal{P}_*[x] \cdot \left(\sum_{l=l_0}^{|I|} \mathcal{P}_{I \setminus \mathcal{G}(f),l-|I \cap \mathcal{G}(f)|}[x] \right) \times d_{x|\bar{e}} + \langle \mathcal{P}_I^{l_0}, \bar{\mathcal{P}}_* \rangle_{d|\bar{e}}$$

so if $l_0 \leq |I \cap \mathcal{G}(f)|$

$$\begin{aligned}
\nu_{I,l_0,f} &= \nu_{\mathcal{G}(f),|\mathcal{G}(f)|,f} + \langle \mathcal{P}_I^{l_0}, \bar{\mathcal{P}}_* \rangle_{d|\bar{e}} \\
&\geq \nu_{\mathcal{G}(f),|\mathcal{G}(f)|,f}
\end{aligned}$$

Since $\langle \mathcal{P}_I^{l_0}, \bar{\mathcal{P}}_* \rangle_{d|\bar{e}} \geq 0$, which lead to a contradiction.

Recall of vertex

Let G be a firing graph with layer 0 composed by measure grid's bits $\mathcal{G}$ and $f \in \mathcal{F}$ denote some target factor that is linked to some bit of the measure grid. The distribution of activation of latent factors and measure grid's bits will be denoted d and d' and the event "factor f is active" will be denoted by e. Let v be a vertex connected to a set I of measure grid's bits with level $l \in \{1, \dots, |I|\}$ and let $f \in \mathcal{F}$ be some factor. The result (7) stands that the recall of v with respect to f is

$$\psi_{v,f} = \mu_{v,f}$$

Furthermore, the result (8) stands that

$$0 \leq \phi_{v,f} \leq 1$$

Where right equality is reached whenever v is connected to a set of measure grid's bit $I \in \mathcal{G}$, with level $l_0 = |I|$ such that $I \subset \mathcal{G}(f)$.

Proof. From the definition of $\psi_{v,f}$ we have

$$\psi_{vf} = \psi_{I,l,f} = \frac{\|\mathcal{P}_I^l\|_{d,e}}{\|\mathcal{P}_{\mathcal{G}(f)}\|_{d,e}}$$

Thus using $\mathcal{P}_* = \mathcal{P}_{\mathcal{G}(f),|\mathcal{G}(f)|}$ one have

$$\psi_{vf} = \frac{\langle \mathcal{P}_I^l, \mathcal{P}_* \rangle_{d,e} + \langle \mathcal{P}_I^l, \bar{\mathcal{P}}_* \rangle_{d,e}}{\|\mathcal{P}_f\|_{d'}}$$

Yet $\langle \mathcal{P}_I^l, \bar{\mathcal{P}}_* \rangle_{d,e} = 0$, thus

$$\psi_{vf} = \frac{\langle \mathcal{P}_I^l, \mathcal{P}_* \rangle_{d,e}}{\|\mathcal{P}_f\|_{d'}} = \frac{\langle \mathcal{P}_I^l, \mathcal{P}_* \rangle_{d|e} \times \|\mathcal{P}_f\|_{d'}}{\|\mathcal{P}_f\|_{d'}} = \mu_{v,f}$$

Finally the result (8) directly comes with the definition.

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vertex's reward process

Let G be a firing graph where the layer 0 is composed by measure grid's bits. v denote some vertex of G and $f \in \mathcal{F}$ denote some target latent factor. We also use d and d' to respectively denote the distribution over combination of activation of measure grid's bits and factors and e to denote the event "factor f is active". If $r_{v,f}[t]$ denote reward process of v with respect to f as defined in section 1.2.2, the result (9) stands that

$$\mathbb{E}\left[r_{v,f}[t]\right] = t \times q_r = t \times (\mu_{v,f} \times \|\mathcal{P}_f\|_{d'})$$

Proof. From the definition of the reward process we have

$$\begin{aligned} r_{v,f}[t] &= r_{v,f}[t-1] + r_{v,f,t} \\ &= \sum_{i=0}^{t-1} r_{v,f,i} \end{aligned}$$

with $\{r_{v,f,i}\}_{i=1,\dots,t}$ a sequence of i.i.d bernouilli trial with parameter $p = q_r = \|\mathcal{P}_v\|_{d,e}$. Thus

$$\begin{aligned} \mathbb{E}\left[r_{v,f}[t]\right] &= \sum_{i=0}^{t-1} \mathbb{E}\left[r_{v,f,i}\right] \\ &= t \times \|\mathcal{P}_v\|_{d,e} \\ &= t \times \langle \mathcal{P}_v, \mathcal{P}_{\mathcal{G}(f),|\mathcal{G}(f)|} \rangle_{d|e} \times \|\mathcal{P}_f\|_{d'} \end{aligned}$$

the proof come with identification of terms.

<u>vertex's penalty process</u>

In the same context, if $p_{v,f}[t]$ denotes penalty process of v with respect to f as defined in section 1.2.2, the result (10) stands that

$$\mathbb{E}\left[p_{v,f}[t]\right] = t \times q_p = t \times (\nu_{I,f} \times (1 - \|\mathcal{P}_f\|_{d'}))$$

Proof. From the definition of the reward process we have

$$\begin{aligned} p_{v,f}[t] &= p_{v,f}[t-1] + p_{v,f,t} \\ &= \sum_{i=0}^{t-1} p_{v,f,i} \end{aligned}$$

with $\{p_{v,f,i}\}_{i=1,\dots,t}$ a sequence of i.i.d bernouilli trial with parameter $p = q_p = \|\mathcal{P}_v\|_d - \|\mathcal{P}_v\|_{d,e}$. Thus

$$\begin{aligned} \mathbb{E}\left[p_{v,f}[t]\right] &= \sum_{i=0}^{t-1} \mathbb{E}\left[p_{v,f,i}\right] \\ &= t \times (\|\mathcal{P}_v\|_d - \|\mathcal{P}_v\|_{d,e}) \\ &= t \times (\|\mathcal{P}_v\|_{d,e} + \|\mathcal{P}_v\|_{d,\bar{e}} - \|\mathcal{P}_v\|_{d,e}) \\ &= t \times \|\mathcal{P}_v\|_{d,\bar{e}} \\ &= t \times \|\mathcal{P}_v\|_{d|\bar{e}} \times \|\|_{d'} \end{aligned}$$

the proof come with identification of terms.

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vertex's score process

Finally, in the same context, if $s_{v,f}[N,T,\rho]$ denotes the score process of v with respect to f as defined in section 1.2.2 with some $N,T,\rho \in \mathbb{N}^3$, the result (11) stands that

$$\mathbb{E}\left[s_{v,f}[N,T,\rho]\right] = N + T \times (q_s \times (\rho+1) - \rho) = N + T \times (\phi_{v,f} \times (\rho+1) - \rho)$$

Proof. From the definition of the score process we have

$$s_{v,f}[N,T,\rho] = N + \sum_{t=1}^{T} s_{v,\rho,f,t}$$

with $\{s_{v,\rho,f,t}\}_{t=1,\dots,T}$ a sequence of i.i.d such that

$$s_{v,\rho,t,f} = \begin{cases} 1, & \text{with probability } q_s = \frac{q_r}{q_r+q_p} \\ -\rho, & \text{with probability } 1-q_s \end{cases}$$

Thus we can write

$$\begin{aligned}
\mathbb{E}\left[s_{v,f}[N,T,\rho]\right. &= N + \sum_{t=1}^{T} \mathbb{E}\left[s_{v,\rho,t,f}\right] \\
&= N + \sum_{t=1}^{T} q_s - \rho \times (1 - qs) \\
&= N + \sum_{t=1}^{T} q_s \times (\rho+1) - \rho \\
&= N + T \times (q_s \times (\rho+1) - \rho)
\end{aligned}$$

Yet $q_s = \frac{q_r}{q_r+q_p}$, with $q_r = \|\mathcal{P}_v\|_{d,e}$ and $q_p = \|\mathcal{P}_v\|_{d,\bar{e}}$, thus

$$q_s = \frac{\|\mathcal{P}_v\|_{d,e}}{\|\mathcal{P}_v\|_{d,e} + \|\mathcal{P}_v\|_{d,\bar{e}}} = \frac{\|\mathcal{P}_v\|_{d,e}}{\|\mathcal{P}_v\|_{d}}$$

Which is the definition of $\phi_{v,f}$.

Furthermore the result (12) stands that,

$$\mathrm{Var}\left[s_{v,\rho,t,f}\right] = (1+\rho)^2 \times \phi_{v,f} \times (1 - \phi_{v,f})$$

Proof. Using the result of previous proof we first compute $\mathbb{E}\left[s^2_{v,\rho,t,f}\right]$

$$
\begin{aligned}
\mathbb{E}\left[s_{v,\rho,t,f}^2\right] &= q_s + \rho^2 \times (1 - q_s) \\
&= \phi_{v,f} + \rho^2 \times (1 - \phi_{v,f}) \\
&= \phi_{v,f} \times (1 + \rho) \times (1 - \rho) + \rho^2
\end{aligned}
$$

Furthermore we have seen previously that $\mathbb{E}\left[s_{v,\rho,t,f}\right] = \phi_{v,f} \times (1 + \rho) - \rho$, thus

$$
\mathbb{E}\left[s_{v,\rho,t,f}\right]^2 = \phi_{v,f} \times (1 + \rho) \times (\phi_{v,f} \times (1 + \rho) - 2 \times \rho) + \rho^2
$$

Finally

$$
\mathrm{Var}\left[s_{v,\rho,t,f}\right] = \mathbb{E}\left[s_{v,\rho,t,f}^2\right] - \mathbb{E}\left[s_{v,\rho,t,f}\right]^2 = (1 + \rho)^2 \times \phi_{v,f} \times (1 - \phi_{v,f})
$$

B Implementation

References

[1] T. Tao. *Topics in Random Matrix Theory, ser. Graduate Studies in Mathematics.* Providence, Rhode Island: American Mathematical Society, 2012.